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Logical Slab Access and Local Processing with Three-Dimensional Layer Codes at Optimal Distance

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Three-dimensional layer codes combine local stabilizer checks with favorable storage parameters. We specify a realization of the Williamson–Baspin construction for Calderbank–Shor–Steane (CSS) inputs with check weights at least two and prove that it preserves the encoded dimension. Every pure-X logical class is accessible in each lattice layer, and every pure-Z class in each transverse plane. For balanced good inputs of length N, these slabs give the optimal distance scaling $\Theta(N^2)$. A three-dimensional gauging grid with constant edge multiplicity preserves static code distance. For the same gauging construction, a bounded-thickness bridge between two blocks instead admits a short logical cut. Finite simulations support this contrast at low noise; separate routed circuits exhibit bridge fault distance one on three inputs and recover the tested idle-memory distances on the smallest blocks with a spanning grid. We give local gate and deterministic resource-distillation constructions and derive space and depth bounds for a universal architecture, conditional on stated uniform detector, composition, and resource-locality hypotheses under local-stochastic Pauli noise.

I. INTRODUCTION

Quantum error-correcting codes must protect stored information and make that information accessible to computation. These demands can pull in different directions: a code with favorable storage parameters may still require costly logical measurements or physical connections that are difficult to realize. Three-dimensional layer codes provide a natural setting in which to address this problem. The construction of Williamson and Baspin embeds a Calderbank–Shor–Steane (CSS) code into a local network of surface-code sheets and defects [1]. For balanced good low-density parity-check inputs of length N, the resulting blocks have

$$n = \Theta(N^3), \qquad K = \Theta(N), \qquad d = \Theta(N^2), \quad (1)$$

with bounded-weight, geometrically local stabilizers. Here n is the physical block size, K its number of logical qubits, and d its distance. A *balanced good* input family has $n_{\mathrm{in}} = N$, $m_X, m_Z = \Theta(N)$, $K = \Theta(N)$, and both input distances $\Theta(N)$, with uniformly bounded row and column weights. The chosen generator lists may be redundant. Geometric spacing is fixed independently of N.

The central result of this work is a geometric interface for processing that stored information. We give an explicit lattice realization of the layer code with explicitly specified check matrices, and show that every pure-X logical class has a representative on a single layer of the lattice and every pure-Z class one in a single transverse plane. The layer can be placed at any height, and the plane at any position, independently of N and of the chosen logical operator. This converts an algebraic choice of logical Pauli into a spatially localized measurement port.

We use this interface to formulate a universal fault-tolerant architecture on the native three-dimensional nearest-neighbor lattice. The argument has three steps. A prefix-parity construction first produces the slab representatives. Relative expansion of a cubic grid then gives a local gauging measurement with distance $\Omega(d)$. Finally, repeated extraction, padded composition, and encoded resource transfer give a complete circuit under local-stochastic Pauli noise, subject to the uniform hypotheses stated in Sec. IV. Figure 1 summarizes the architecture.

Logical measurement by gauging, surgery, and homological constructions has become a general route to processing quantum LDPC codes [2–6]. Adapters, extractors, and universal architectures address complementary questions of connectivity, parallelism, and resource overhead [7–10]. Here the organizing constraint is three-dimensional Euclidean locality: logical access, resource delivery, and syndrome extraction use the same nearest-neighbor geometry. The layer-code slab theorem supplies the missing connection between the memory construction and these processing primitives.

Locality also fixes the achievable overhead. In three dimensions a local stabilizer code obeys $Kd = O(n)$ [11], so $\Theta(d)$ physical qubits per logical qubit is the best possible storage cost, and layer codes attain it. Surface-code patches processed by lattice surgery use $\Theta(d^2)$ qubits per logical qubit [12, 13], and three-dimensional surface and color codes trade larger volume for transversal gates [14]. Local simulations of LDPC codes [15], local fault-tolerance constructions [16], wire codes [17], and planar logical measurements [18] address related geometric questions, while constant-overhead schemes [19–21] and LDPC magic-state injection [22, 23] rely on nonlocal connectivity. Under its stated uniform hypotheses, the architecture below keeps the $\Theta(d)$-per-logical storage cost during computation, up to the cost of one encoded cell for bounded-aspect logical boxes; general regions retain a partial-occupancy correction.

This report proves the slab and static-distance results, specifies the local logical primitives, and states separately the additional hypotheses needed for the architecture bound. Section VIII specifies the lattice and numerical protocol; Appendices A–E give supporting proofs and the precise conditional resource accounting. Finite computations test the construction and interfaces; they do not establish uniform properties of arbitrary adaptive schedules. Slab mobility also gives the port-packing and

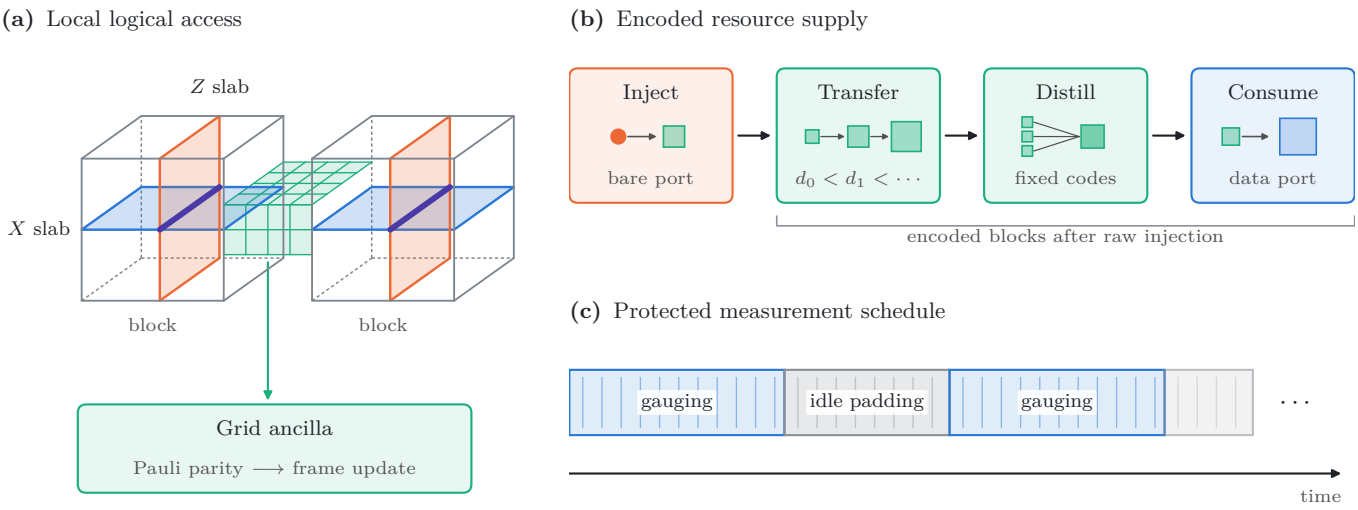


FIG. 1. Geometry-preserving processing. (a) A layer-code block exposes transverse slab ports for pure-X and pure-Z logical operators. A local grid ancilla measures the selected Pauli; neighboring blocks use the same interface for joint measurements. (b) A raw resource is injected into an encoded block, transferred through blocks of increasing distance, distilled, and consumed by the data path. (c) Repeated extraction and idle padding connect successive active measurements. All drawings are schematic; the encoded blocks and grid ancillas occupy three-dimensional regions.

shallow-unitary results of Sec. V.

II. LOGICAL OPERATORS IN CONSTANT-THICKNESS SLABS

A. The layer code as glued sheets

A layer code is assembled from surface-code sheets [1]. Each qubit q of the input CSS code becomes a *data sheet*, the plane $y = Y_q$; each input Z check becomes a Z-check sheet, a plane $x = \text{const}$; and each input X check becomes an X-check sheet, a plane $z = \text{const}$. Every sheet is an ordinary surface code on a rectangle of the cubic lattice, so it carries string operators. An X string joining the two boundary components on which its endpoints condense commutes with all checks of that sheet; a string that stops inside the sheet leaves a violated check, a *defect*, at its end. A Z-check sheet meets the data sheets of the qubits in its check along vertical lines, the *junctions*. Ordered along the check, these are its first, middle, and last junctions. The X-check sheets are attached in the same way, with X and Z exchanged.

A junction is a rule that says which strings may end on the line without leaving a defect. For the X strings used below, only their presence matters. At a first or last junction the Z-check sheet ends on the data sheet, and a string may pass from one sheet to the other: the check-sheet string is present exactly when the data string is. At a middle junction three strings meet the line: the data string, which crosses it, and the check-sheet strings on its two sides. They leave no defect exactly when an even number of the three is present. Writing 1 for a present string, and listing the check-sheet strings first (left, then right) and the data string last,

$$H_{\text{end}} = \begin{bmatrix} 1 & 1 \end{bmatrix}, \qquad \ker H_{\text{end}} = \{00, 11\},$$
$$H_{\text{mid}} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}, \quad \ker H_{\text{mid}} = \{000, 110, 101, 011\}. \quad (2)$$

Williamson and Baspin state these rules as sets of condensed anyons, that is, as the combinations of string endpoints that disappear at the line; Eq. (2) is the same information written for X strings. The word 110 permits the pass-through needed below. It lets a string on the check sheet pass a data sheet without any string on that data sheet. If a middle junction allowed only 000 and 111, every check-sheet string would drag a string onto each data sheet it passes, and a general logical operator could not be built; such a lattice in fact fails to encode the input qubits (Table II).

B. An explicit lattice for the junctions

We build every junction on the lattice (Sec. VIII A). Sheets own distinct qubits, and each junction rule adds one qubit of one sheet to one check of the other. At a middle junction the Z-check-sheet face beside the line gains the data sheet's edge on the line. That face then contains three qubits of interest: the check-sheet edge on the line, where the string from one side ends; the next check-sheet edge, where the string on the other side begins; and the added data edge. Its check is therefore exactly the middle rule of Eq. (2). The data stars on the line gain the check sheet's edges on the same side, which keeps them commuting with the modified faces. At a first or last junction the check sheet ends on the data line and shares its edges, which gives the two-bit rule.

One more repair is needed. An X-check sheet and a Z-check sheet cross along a line, and there the junction additions of the two sheets make a few stars and faces anticommute, once for each data qubit shared by the two input checks. A row of one-qubit additions along the crossing line, which we call a *wall*, removes these conflicts. It exists exactly when the shared number is even, which is the commutation of the two input checks. The wall additions do not alter the slab commutation tests: the added a_y terms in Z checks are absent from X slabs, and the added c_z terms in X checks are absent from Z slabs. Every check lies within unit distance of one lattice vertex and has weight at most six. The junctions have the condensed anyons of Ref. [1]; we do not claim that the lattice is identical to the drawings there, and all results below are proved for this explicit lattice, specified in Sec. VIII A.

C. Slab representatives

Theorem II.1 (Uniform slab representatives)**.** *For every input CSS code with checks of weight at least two, the explicit realization encodes exactly K logical qubits. Every pure-X logical class has a representative supported on the single layer of z-edges spanning $[z_0, z_0+1]$, for every height $z_0 \in \{0,\dots,L_z-1\}$ of the block, and every pure-Z class has a representative in the plane $x = x_0$, for every $x_0 \in \{0,\dots,L_x\}$. The heights and planes are chosen independently of the logical classes.*

The operator is built at one fixed height z_0, one input check at a time. Choose $x \in \ker H_Z$ of the input code, representing a pure-X class modulo $\mathrm{row}(H_X)$, and put an X string at height z_0 on the data sheet of every qubit with $x_q = 1$. Each such string crosses the junction of every Z-check sheet attached to its data sheet, and on its own would leave a defect there. Along one Z-check sheet the selected junctions must therefore be joined, in pairs, by strings on the check sheet at the same height; we call these check-sheet strings the *backbone*. Pairing is possible because x commutes with the input check, so the check contains an even number of selected qubits. Walk along the check from its first qubit and switch the backbone on or off at every selected junction. With ordered incidences $x_1,\dots,x_k$, the backbone after the ith junction is present exactly when an odd number of selected qubits lies behind it, a running count that we call the *prefix parity*,

$$p_0 = 0, \qquad p_i = \sum_{h=1}^{i} x_h, \qquad p_k = 0, \tag{3}$$

where $p_k = 0$ is the commutation $\sum_i x_i = 0$ over $\mathbb{F}_2$. At every middle junction $p_{i-1} + p_i + x_i = 0$, so none or two of the three strings are present, as Eq. (2) requires; at the ends $p_1 = x_1$ and $p_{k-1} = x_k$. No defect remains anywhere (Fig. 2). Every piece lies in the same layer of vertical edges, so the operator is one lattice layer thick, whatever the block size and whichever logical is chosen. Building it at another height gives another representative of the same class. In the explicit realization the check is exact and local: every stabilizer meets the operator in one of the forms $x_1 + p_1$, $p_{i-1} + p_i + x_i$, $p_{k-1} + x_k$ of a single incidence, or not at all.

The pure-Z operators are built in the same way in a vertical plane $x = x_0$, with backbones on the X-check sheets. An X slab operator and a Z slab operator share only one data edge for each input qubit selected by both, so they anticommute exactly when the two input logicals do. This gives K independent logical qubits. A count shows that there are no others. On its own, each data sheet would encode one qubit and each check sheet would remove one; the junctions and walls add neither qubits nor checks; and every redundancy among the lattice checks comes from a redundancy among the input checks. In formulas, each data sheet contributes $+1$ and each check sheet -1 to $n - \#X - \#Z$, and the relations among the lattice checks inject into those among the input checks. Hence the realization encodes at most, and therefore exactly, K qubits, and the slab operators represent every pure-type class. Appendix A gives the relation maps and proves the logical-space isomorphism, including redundant input checks and translation of the slabs. Finite certificates on complete codes with up to 10,936 qubits verify these properties on the tested instances (Sec. VIII B and Table II).

Williamson and Baspin already obtain the layer-code logical operators by quasi-concatenating input logicals [1]. The new ingredients are an explicit realization in which this construction can be verified check by check, a single-layer thickness independent of the block size and of the chosen logical, and its uniformity over every height, which supplies the disjoint ports used below.

The same ports give the distance, by a simple idea: a logical operator must be visible in every plane that contains a complete port of the other type. Let O be a nontrivial X logical. It anticommutes with some Z logical, and that Z logical has a slab representative in each of the $L_x + 1$ disjoint planes $x = x_0$, so O must place qubits in every one of these planes. To count them, cut the Z slabs of one plane into their pieces: a vertical data column for each input qubit, and the backbone segments. Record, for each piece, whether O meets it an odd number of times. Read a data piece as its input qubit, and a backbone piece as the qubits of its input X check that lie behind it, the first t of them for the tth segment. The resulting vector $\pi_{x_0}(O)$ commutes with every input Z stabilizer because its pairing equals that of O with the corresponding lifted stabilizer. It pairs nontrivially with a conjugate input logical, whose lift can be placed in this plane. Thus it is a nontrivial class in $\ker H_Z / \mathrm{row}(H_X)$ and has at least d_X^{in} entries (Appendix A). The first t qubits of a check may be exchanged for the remaining ones, because the whole check is a stabilizer of the input code, so a backbone piece stands for at most $\eta_X = \max\{1, \lfloor w_X/2 \rfloor\}$ qubits, where w_X is the largest input X-check weight. Each piece met

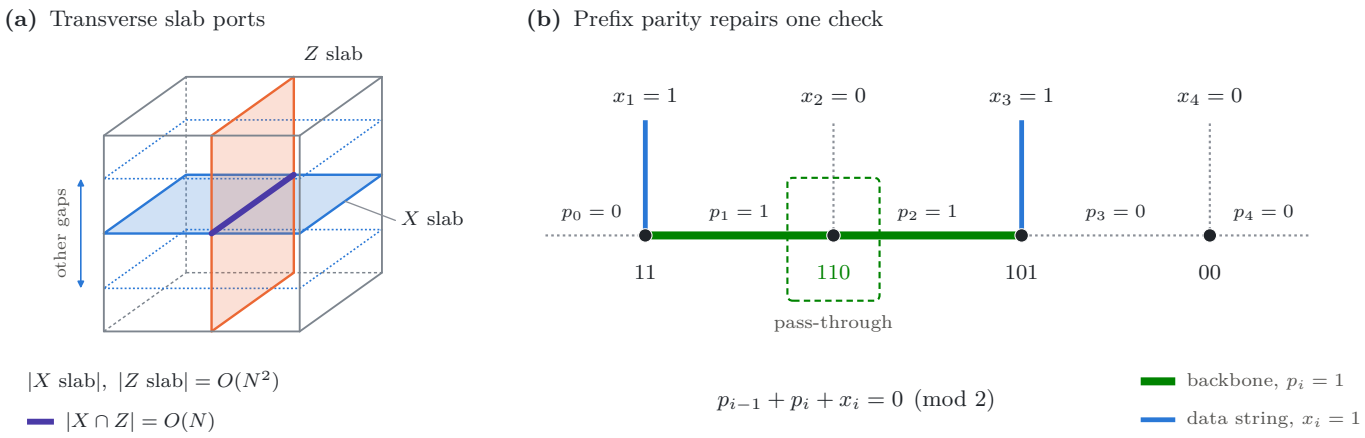


FIG. 2. Slab mobility and its local mechanism. (a) Complete X and Z ports occupy transverse single-layer slabs. Each slab contains $O(N^2)$ qubits, and their intersection contains $O(N)$ qubits. Translating to another height (or plane) gives another representative of the same logical class. (b) For the even incidence word $(1,0,1,0)$, the running parity is $(0,1,1,0,0)$. The highlighted middle cell uses 110: the backbone remains occupied although the data incidence is absent. Endpoint cells use the two-bit relation in Eq. (2).

an odd number of times holds at least one qubit of O, so each plane contains at least $\lceil d_X^{\text{in}}/\eta_X \rceil$ qubits of O, and

$$d_X \geq (L_x + 1) \left\lceil \frac{d_X^{\text{in}}}{\eta_X} \right\rceil , \tag{4}$$

and, with $\eta_Z = \max\{1, \lfloor w_Z/2 \rfloor\}$,

$$d_Z \geq L_z \left\lceil \frac{d_Z^{\text{in}}}{\eta_Z} \right\rceil . \tag{5}$$

For balanced good inputs the realization therefore has $n = \Theta(N^3)$, $K = \Theta(N)$, and $d = \Theta(N^2)$, the parameters of Eq. (1). For the $[[4,2,2]]$ and $[[6,4,2]]$ inputs the bound equals the exact distance in every computed instance.

For balanced inputs the block has $\Theta(N)$ heights and $\Theta(N)$ transverse positions, so there are

$$r_X = \Theta(N), \qquad r_Z = \Theta(N) \tag{6}$$

pairwise-disjoint slab regions within each orientation. Each region is a complete port for its logical polarization. This repeated access will also give the structural bounds in Sec. V.

III. LOCAL GAUGING AND UNIVERSAL LOGICAL OPERATIONS

A. Why a three-dimensional grid preserves distance

Gauging measures a logical operator by measuring many small checks whose product is that operator [2]. Let $\overline{P} = X(D)$ be the selected logical representative. Attach each data qubit of D to its own vertex of a connected grid $G = (V, E)$, call these vertices the *ports* U, and place an ancilla qubit on every edge. The new check at a vertex, its *vertex star*, is X on the edges at that vertex and on the attached data qubit, if any. Every edge belongs to two vertex stars, so the product of all vertex stars is $\overline{P}$, and the product of their measured outcomes is the logical outcome; yet each measurement involves only a few neighboring qubits. The grid also carries Z checks around its unit squares, and every original Z check that meets D is extended along short grid paths joining its ports, so that it commutes with the stars. Preparing the edge qubits in $|0\rangle$, measuring the new checks, and finally reading the edges in Z returns the data to its original code with an updated Pauli frame. The X/Z-exchanged construction measures pure-Z logicals.

The danger is that the added qubits create new, shorter logical operators. The quantity that controls this is the expansion of the grid relative to its ports,

$$\beta(G,U) = \min_{\substack{S \subseteq V \\ 0 < |S \cap U| < |U|}} \frac{|\delta S|}{\min\{|S \cap U|, |U \setminus S|\}}, \tag{7}$$

where δS is the set of edges leaving S and only positive denominators are included; throughout this lemma $|U| \geq 2$. We call δS a *wall*: it separates the ports in S from the others, and β is the smallest number of wall edges paid per port separated. For a code with CSS distances d_X, d_Z, the deformed distances satisfy

$$\widetilde{d}_X \geq \min\{1, \beta(G,U)\} d_X, \qquad \widetilde{d}_Z \geq d_Z. \tag{8}$$

The reason is short. An X logical of the deformed code may put part of its weight on the ancilla edges, but those edges must form a wall δS; any other edge set anticommutes with some square Z check. Multiplying by the vertex stars in S removes the wall and puts X instead on the data qubits attached in S, or, using the complement of S, on those attached outside it. Either way the result is a logical of the original code, so it has

weight at least d_X. The trade removes $|\delta S|$ edges and, on the better side, adds at most $m = \min\{|S \cap U|, |U \setminus S|\}$ data qubits, while $|\delta S| \geq \beta m$. Hence the deformed logical weighs at least $\min\{1, \beta\}$ times a logical of the original code. If $\beta \geq 1$, no logical saves weight by moving into the ancilla. For the Z sector, restriction to the original data maps a nontrivial deformed logical class to a nontrivial original logical class. Indeed, if the restriction is an original stabilizer, its deformed lift removes that data support; the remaining ancilla support is a cycle and hence a stabilizer. Ancilla support can be necessary for commutation, but cannot reduce the weight of the data restriction. This proves $\widetilde{d}_Z \geq d_Z$.

The slab geometry makes β a constant, and this is where three dimensions matter. A slab representative lies in one plane and has $\Theta(N^2)$ qubits, hence $\Theta(N^2)$ ports. If the grid were only that plane, a straight line across it would wall off half of the ports with $\Theta(N)$ edges, so $\beta = O(1/N)$. If the grid is a box as thick as it is wide, a wall must be a surface. Either it separates roughly the same region in every layer of the box, and then it cuts through all the layers, or the separated region changes from layer to layer, and every change costs edges between layers. In both cases its area grows in proportion to the number of ports it separates. If U occupies at most t parallel planes of a box with aspect ratio at most A, Appendix B proves

$$\beta(G, U) \geq \frac{1}{2At}. \tag{9}$$

Constant refinement assigns distinct local port labels to distinct grid vertices and increases t only by a constant. Unit-square cycles, bounded pairing paths, and bounded local density keep the check weight, diameter, and incidence constant. Thus one block or two blocks at fixed bounded cell separation admit local logical measurement using $O(n)$ ancillas and retaining distance at least $\beta_0 d$ for a fixed $\beta_0 > 0$. Section III B shows that $\beta_0 = 1$ is attainable.

B. Keeping the distance throughout the computation

A logical measurement can lower the fault distance, the smallest number of faults that cause an undetected logical error, in three ways. The added qubits can carry a short logical operator; a few wrong measurement outcomes can flip the reported result; and a few faults can exploit the join between two consecutive operations. Figure 3 shows how the construction blocks each of them.

The first danger is real for thin interfaces. Consider a joint measurement $\overline{X}_a\overline{X}_b$ of two blocks. Once it has been measured, $\overline{X}_a\overline{X}_b$ is a stabilizer, so $\overline{X}_a$ and $\overline{X}_b$ act in the same way on the code. Take any wall δT with the ports of block a on one side and those of block b on the other. The product of the vertex stars in T is $\overline{X}_a$ times X on the wall, so $X(\delta T)$ alone is a logical operator equivalent to $\overline{X}_a$ [Fig. 3(a)], and

$$\widetilde{d}_X \leq \min\{|\delta T| : \ T \cap U = U_a\}. \tag{10}$$

For layer-code blocks of side $\Theta(N)$, the distance $d = \Theta(N^2)$ is comparable to the area of a block face. For these joint gauging measurements, assume bounded local qubit density and edge multiplicity and a bridge of thickness h and transverse extent $O(N)$. A separating plane meets the bridge in $O(hN)$ edges. If $\overline{X}_a$ survives as a nontrivial class after measuring the product, Eq. (10) bounds the deformed static distance by $O(hN)$. This is a cut obstruction for the specified gauging graph. Joint measurements cannot be avoided: they implement CNOT, phase teleportation, and resource transfer. A complete circuit-level simulation on explicit layer-code blocks (Sec. VI) finds the same effect in the fault distance of the measurement circuit, exactly at two block sizes and, with routed nearest-neighbor extraction, for three input codes, and in its failure rate at four sizes.

Four ingredients keep the distance. First, the grid fills the participating blocks, so every wall between their ports is a surface with $\Omega(N^2)$ edges [Fig. 3(b)]; this is the constant relative expansion of Eq. (9), which the constant slab thickness makes possible. Second, carrying $m_0 = \lceil 2At \rceil$ parallel ancillas on every grid edge, with the square cycles on designated copies and the weight-two parallel-edge cycles of Appendix B, multiplies every wall by m_0, so $\beta \geq 1$ and Eq. (8) gives $\widetilde{d}_X \geq d_X$: each measurement keeps the full static distance at constant ancilla cost. Third, the measurement is repeated in time [Fig. 3(c)]. A single wrong outcome of one vertex star flips the reported product, so the new checks are measured for $R = \Theta(d)$ rounds, and a wrong outcome that is not repeated is caught by comparing consecutive rounds. In the measurement-only example, flipping one star record in all R rounds changes the reported product without a consecutive-round detector. Mixed data and record faults require a slice argument, rather than the assumption that one fixed outcome is wrong in every round. In the elementary phenomenological model a fault is one single-qubit Pauli or one record flip. For a relative gauging rectangle with zero input error and zero output syndrome and records, Appendix C proves

$$f_{\mathrm{ph}}^{\mathrm{rect}} \geq \min\{\widetilde{d}_X, d_Z, R\}. \tag{11}$$

The argument retains arbitrary inherited error data by subtracting their fault-free propagation. If each physical circuit fault has an equivalent image of at most c_1 elementary mechanisms, then

$$f_{\mathrm{circ}}^{\mathrm{rect}} \geq \left\lceil f_{\mathrm{ph}}^{\mathrm{rect}}/c_1 \right\rceil. \tag{12}$$

Thus static-distance preservation and circuit fault-distance preservation are distinct statements. With bounded spread the resulting circuit lower bound is $\Omega(d)$ when $R = \Theta(d)$. Fourth, padding successive measurements by $\Theta(d)$ idle rounds supports composition under the explicit conditions in Sec. IV.

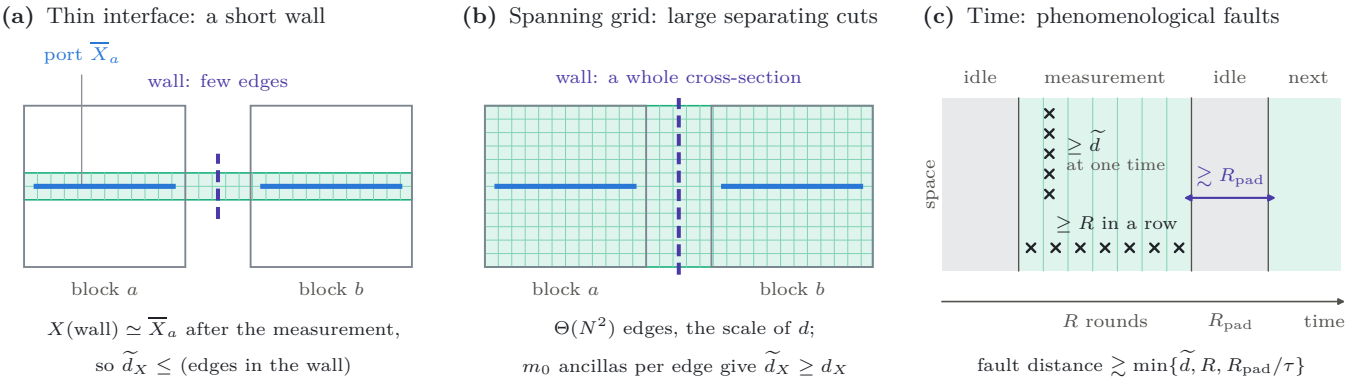


FIG. 3. Spatial and temporal mechanisms for distance protection. (a) For joint gauging with a thin separating bridge, the cut operator is equivalent to the surviving $\overline{X}_a$ class and bounds the static distance. (b) Constant relative expansion and the complete cycle checks of the amplified grid preserve static distance. (c) In the elementary phenomenological model, the slice argument bounds detector-silent protected faults by $\min\{\widetilde{d}_X, d_Z, R\}$. A measurement-only example flips one star record in all R rounds. General circuit faults require the bounded-spread conversion, and composition requires the hypotheses in Sec. IV. The drawing is schematic.

Resource transfer joins blocks of unequal distance. Let C_1 be the smaller block and C_2 have distance at least d_2. For a joint ZZ gauging measurement define $\mathcal{Q}_1$ as the subgroup of deformed logical classes induced by logical operators of C_1 commuting with its measured $\overline{Z}_1$; representatives are completed along grid edges when required. The measured product identifies $\overline{Z}_1$ with $\overline{Z}_2$ in this subgroup. The minimum weight of a class outside $\mathcal{Q}_1$ is at least $\min\{1, \beta\}d_2$ (Appendix B). This is a quotient distance, not a lower bound on the entire two-block code. Dedicated courier blocks carry one resource, with unused logical qubits in fixed states and subsequently discarded. Protection of retained spectators during noisy transfer also needs the relative propagation condition stated below.

C. Gates and encoded resources

Pure-X/Z measurements suffice for the logical gate set. A CNOT uses an ancilla a in $|+\rangle$ and the measurements

$$Z_c Z_a \ (u), \qquad X_a X_t \ (v), \qquad Z_a \ (w), \tag{13}$$

followed by the Pauli-frame update $Z_c^v X_t^{u \oplus w}$. Outcome bit zero denotes eigenvalue $+1$.

For phase teleportation, prepare a in $R_\theta|+\rangle$, where $R_\theta = \mathrm{diag}(1, e^{i\theta})$, measure $Z_d Z_a$ with bit m and X_d with bit n, and retain a. The correction

$$C_{mn} = R_{2\theta}^m Z_a^n X_a^m \tag{14}$$

produces $R_\theta|\psi\rangle$. For $\theta = \pi/2$ this consumes $|Y\rangle = S|+\rangle$; for $\theta = \pi/4$ it consumes $|A\rangle = T|+\rangle$, with a possible S correction using one additional $|Y\rangle$. Exchanging X and Z gives $S_X = HSH$, and $SS_XS = e^{i\pi/4}H$. These identities generate Clifford+T using the one- and two-block measurements above.

A physical resource enters at a designated bare port. The raw qubit is teleported into an encoded block by a joint ZZ measurement and an X readout. Its weight-one error class is the intentional resource degree of freedom; the static spectator quotient retains distance $\Omega(d)$. The corresponding circuit statement uses the relative-transfer hypothesis in Sec. IV. Subsequent transfers use only encoded blocks, with distances growing by fixed factors. We call this sequence an encoded courier. Each courier block is dedicated to one traveling resource; its other logical qubits are prepared in fixed states and never read.

The resource supply uses fixed Steane $7 \to 1$ and punctured Reed–Muller $15 \to 1$ factories, implemented by the same logical primitives [24–26]. Resource errors are modeled as local-stochastic defective slots. For any set J of preassigned candidate slots, the probability that all are defective is at most $q_r^{|J|}$. Conditional on a defect set, good slots are ideal product resource states; bad slots can be arbitrary and entangled with one another or with an environment. This is a joint condition, not merely a bound on each marginal error probability.

For phase teleportation with a defective resource, every conditional Kraus operator, relative to the ideal rotation, lies in $\mathrm{span}\{I, Z\}$. With an explicit environment the branch map has form $I \otimes E_0 + Z \otimes E_1$. The fixed recovery in Appendix D therefore returns the ideal output whenever at most one coordinate of a run is defective. Each run always produces an output, and disjoint runs obey the family bound

$$q_{r+1} \le C_{\text{dist}} q_r^2, \qquad C_{\text{dist}} = \binom{30}{2} = 435. \tag{15}$$

The 30 preassigned slots are fifteen principal candidates and fifteen possible correction candidates. Unused candidates are discarded. The one-defect property and the joint bound give a union bound over pairs for each disjoint output run, without assuming independence or twirling. Equation (15) treats all other logical operations as ideal;

their physical faults must be covered separately by the architecture hypotheses.

IV. FAULT TOLERANCE AND RESOURCE SCALING

A. From local distance to a complete circuit

We consider local-stochastic Pauli circuit noise: any specified set F_0 of mechanisms at distinct circuit locations occurs with probability at most $p^{|F_0|}$. A full architecture needs uniform properties of the scheduled detector and effect maps, beyond the static-distance lemmas and finite circuit checks. We state those properties explicitly.

At a temporal cut use (s, λ, ρ) for the syndrome, logical Pauli class, and last-round record error, all relative to a fixed ideal branch and chosen Pauli representatives. These describe an error class, not an arbitrary quantum state. For a fixed Clifford branch, the detector and effect maps are binary linear maps on these variables and the internal fault vector. Composition must carry all three variables; a faulty record can change ρ without changing s or λ. Changing logical representatives between cuts can add a term linear in s to the propagated frame. The relative effect of an internal fault vector subtracts the fault-free propagation of the same incoming variables.

Assumption IV.1 (Uniform scheduled-circuit hypotheses)**.** There is a nearest-neighbor implementation with a fixed causal skeleton and a finite set of adaptive branch templates satisfying the following properties uniformly in block size, duration, and realized history.

1. A round has bounded depth and circuit-to-phenomenological spread c_1. In the union of the branch detector catalogs, each detector involves at most a elementary mechanisms, each mechanism influences at most b detectors, and at most a_{loc} mechanisms share a location. Edges joining mechanisms in one detector or location have temporal range at most τ. All these constants are size independent.

2. Boundary maps compose exactly for arbitrary inherited (s, λ, ρ). A detector-silent component contained in an active or idle window, with cuts in fault-free padding, has zero syndrome and record error at its cuts. Its relative effect is its global contribution. Relative distances obey $\delta_g^{\mathrm{rel}} \geq \min\{\beta_0 d, R_g\}$ and $\delta_{\mathrm{mem}}^{\mathrm{rel}} \geq d$. At transfer boundaries, inherited errors on a courier, including its syndrome and record terms, affect only its traveling resource and discarded logical qubits; retained spectators obey the corresponding large-block quotient bound.

3. Each retained block participates in at most one active window at a time, with R_{pad} idle rounds between windows. A detector belongs to one idle block or one active primitive; no detector joins distinct lifetimes of a reused cell. Courier blocks are dedicated to one resource. Distinct resource neighborhoods have union-graph separation $\Omega(R)$.

4. A committed minimum-cardinality decoder with a $\Theta(d)$ buffer has the following witness property. A wrong protected commitment gives a connected set of $\Omega(d)$ mechanisms with a fixed positive fraction of actual faults. Except on a large-cluster event bounded by $CV(C'p^{1/4})^{cd}$, wrong resource commitments have mutually disjoint connected half-dense witnesses. Here V is the total number of circuit locations. A witness first created at courier stage j has size at least γd_j and an anchor set of size at most $Cd_j^{5/2}$; the raw stage has constant anchor size and witness size at least one. Terminal anchor size is $O(f^{5/2})$, with witness size at least γd_J.

The relative distances refer to nonzero protected effect with zero detectors and zero relative cut syndrome and records. They do not assign distance d to an unrestricted open cut: one late data fault can change its syndrome. Appendix C proves the elementary rectangle bound under these boundary conditions. The uniform extension to arbitrary adaptive histories, including the spectator propagation and decoder-witness properties, is an additional hypothesis of the architecture result. Finite schedule and single-fault checks do not discharge it.

Under Assumption IV.1, a detector-silent vector splits into detector-silent connected components: all support vertices incident on one detector are adjacent, so different components cannot cancel its parity. Effects add on each fixed branch. A harmful component contained in one window pays its relative distance. A component meeting successive primitives on one block must cross the intervening idle pad. Consequently

$$f_{\mathrm{ph}} \geq \min\left\{\min_g \delta_g^{\mathrm{rel}}, \delta_{\mathrm{mem}}^{\mathrm{rel}}, 1 + \left\lfloor \frac{R_{\mathrm{pad}} - \tau}{\tau} \right\rfloor \right\}. \quad (16)$$

The bound is used with $R_{\mathrm{pad}} > \tau$ and complete idle rounds at the cuts. A component shorter than the padding term fits within one such relative rectangle; the fault-free trailing round sets its cut records and syndrome to zero. Choosing all full-distance windows and buffers of order d gives $\Omega(d)$ circuit distance after division by c_1.

Both incidence bounds in item 1 are necessary. They give fault-graph degree at most $b(a-1)+a_{\mathrm{loc}}-1$; bounded detector support alone would not. For fixed maximum degree, connected-set counting and the witness property give exponentially small protected-failure probability. A geometrically growing courier sequence gives a joint injected-resource parameter $q_0 = O(\sqrt{p})$ by the charging estimate in Appendix E. Thus the following theorem is a conditional implication with explicit uniform hypotheses, rather than an inference from finite simulations. Efficient classical decoding is not assumed.

B. Architecture theorem

The input family is a balanced good CSS LDPC family with boundedly dense available sizes: every sufficiently large target length has an available member within a fixed multiplicative factor. Fixed-degree Quantum Tanner families provide the existential input used here [27, 28]. Let $A \subset \mathbb{Z}^3$ be the finite allocated logical region, $W = |A|$, and T the logical circuit depth. Allocated working sites are included in W; physical gate ancillas and live factory resources are included in the bound below.

To state the geometric overhead, partition the logical lattice into coarse boxes of side $b(f) = \Theta(f^{1/6})$. Define $B_f(A)$ as the number of boxes that meet A but are not completely filled. This counts partial occupancy at the block-packing scale, including boundaries, holes, and fragmentation.

Theorem IV.2 (Conditional universal architecture)**.** *Use the explicit layer-code realization of Theorem II.1, a balanced good input family with boundedly dense sizes, and the local measurement, resource, and extraction schedules satisfying Assumption IV.1. There are constants $p_{\rm th}, c, f_{\min} > 0$ such that for $0 < p < p_{\rm th}$ and target failure $0 < \varepsilon < 1/2$, any finite nearest-neighbor three-dimensional logical circuit over Clifford+T, Pauli preparations and measurements, with $W, T \geq 1$, can be simulated with*

$$f = \max\left\{ f_{\min}, \left\lceil \frac{c \log(WT/\varepsilon)}{\log(p_{\rm th}/p)} \right\rceil \right\}, \tag{17}$$

using

$$Q_{\rm phys} = O\Big(Wf + (B_f(A) + 1) f^{3/2} \Big), \tag{18}$$

$$D_{\rm phys} = O(Tf^{3/2}\Psi), \qquad \Psi = O(f^{\log_2 15}). \tag{19}$$

Here W and T are measured after compilation into this finite gate set; ε bounds implementation failure. If synthesis of other gates is required, its approximation error and compilation overhead must be allocated separately. Persistent data remain in layer-code blocks, and physical operations use nearest-neighbor connectivity. For a connected bounded-aspect logical box,

$$Q_{\rm phys} = O(Wf + f^{3/2}). \tag{20}$$

The scaling follows directly from the code parameters. Choose $d = \Theta(f)$ and measurement, padding, and commitment buffers of order f. Equation (1) then gives

$$N = \Theta(\sqrt{f}), \qquad K = \Theta(\sqrt{f}), \qquad n = \Theta(f^{3/2}). \tag{21}$$

A full coarse logical box fits into one cell. Counting full and partial boxes requires $O(W/K + B_f(A) + 1)$ data/factory cells. Multiplication by n proves Eq. (18). Neighboring logical sites map to the same cell or adjacent cells, and a bounded coloring schedules inter-cell gates.

For a bounded-aspect logical box, $B_f(A) = O((W/\sqrt{f})^{2/3} + 1)$, so the partial-cell term is absorbed into Eq. (20). The one-cell term dominates for $W \ll \sqrt{f}$; the extensive term dominates for $W \gg \sqrt{f}$. Thin or fragmented regions retain the general correction.

For time, a distilled resource takes $O(f\Psi)$ rounds, where ℓ distillation levels give $\Psi = O(15^\ell)$ and $2^\ell = O(f)$. One logical layer requires $O(K)$ resources per data block, and serial production gives $O(Kf\Psi) = O(f^{3/2}\Psi)$ rounds per layer, that is $O(f^{3/2+\log_2 15}) \subseteq O(f^{5.41})$. This serial schedule is chosen to keep the space bound; parallel factories would trade space for depth. A depth-first factory schedule stores $O(\ell)$ candidates at once, which fits within the reserved logical capacity. The complete location count is polynomial in f times $WT\Psi$; the exponential cluster bound and final-resource union bound are both at most $\varepsilon/2$ for the choice in Eq. (17).

The depth deserves as much attention as the space. The serial schedule takes $O(Kf\Psi)$ rounds per logical layer, so the storage advantage alone does not establish a practical space–time advantage. In a bounded-aspect logical box, $1 \leq P \leq K$ independent resource lanes per data cell can, under the same throughput and separation hypotheses, give

$$\begin{aligned} Q_{\rm phys} &= O\big(P(Wf + f^{3/2})\big), \\ D_{\rm phys} &= O\big(Tf\Psi\lceil K/P \rceil\big). \end{aligned} \tag{22}$$

The factor P includes factory cells even for a partially occupied data cell. The ceiling retains a resource-latency floor. For $P \leq K$ this gives the same order of upper bound on qubit-time, $O(Tf\Psi K(Wf + f^{3/2}))$; upper bounds do not imply equality of actual costs. Hardware comparisons must include constants, routing, and decoder costs. The architecture implication extends to another code family only if it satisfies both the stated geometric properties and Assumption IV.1.

V. PORT PACKING AND SHALLOW LOGICAL GATES

Slab mobility also constrains how logical information can be spatially accessible. For a region A, let $g_X(A)$ be the dimension of the pure-X logical space supported in A, and define $g_Z(B)$ analogously. The rank $r(A, B)$ of their commutation pairing obeys

$$\max\{0, g_X(A) + g_Z(B) - K\} \leq r(A, B) \leq |A \cap B|. \tag{23}$$

The upper bound follows because all cross commutation factors through the intersection; the lower bound follows from nondegeneracy of the full logical X/Z pairing.

If $A_1, \ldots, A_{r_X}$ are disjoint complete X ports and $B_1, \ldots, B_{r_Z}$ disjoint complete Z ports, summing Eq. (23) over all pairs gives the crossbar bound

$$r_X r_Z K \leq n. \tag{24}$$

Every crossing carries rank K, and the crossings are pairwise disjoint. For the balanced layer-code slabs,

$r_X, r_Z = \Theta(N)$, $K = \Theta(N)$, and $n = \Theta(N^3)$, so the bound is saturated up to constants. Each transverse intersection has size $\Theta(K)$, while each complete slab has size $\Theta(d)$. Thus the cubic physical size is optimal among CSS families required to provide $\Theta(K)$ disjoint complete ports in both polarizations. This is a bound on simultaneous spatial access, distinct from a storage lower bound.

There is a complementary constraint on unitary processing. For a logical Pauli $P = X(x)Z(z)$, pair the ith slab representative of $X(x)$ with the ith representative of $Z(z)$. Each qubit belongs to at most two resulting representatives, giving disjointness $\Delta = \Omega(N)$. The logical-weight bounds sharpen this to

$$\Delta = \Theta(N), \qquad d_\downarrow = \Theta(N^2), \qquad d_\uparrow = \Theta(N^2), \quad (25)$$

where $d_\downarrow$ and $d_\uparrow$ are the smallest and largest minimum weights among nontrivial logical Pauli classes. The Jochym-O'Connor–Kubica–Yoder criterion then forces every fixed-depth, bounded-arity, code-preserving unitary on a sufficiently large static block to be logical Clifford [29]. The result holds even without geometric locality of those gates. The architecture uses adaptive measurements, resource states, and changing codes, which lie outside that unitary setting.

VI. FINITE NUMERICAL EVIDENCE

The supplied simulations test finite interfaces in the construction. The layer-code campaigns below use explicit blocks; the circuit-level campaigns with a clean crossing use surface codes. The largest campaign follows a single encoded planar surface code of distance L, with $n = L^2 + (L-1)^2$ data qubits, through an algebraic base-code → gauged-strip → base-code circuit, without resetting the data between stages. The gauged strip is a path of L ancilla qubits attached to a weight-L logical representative, so the campaign tests the gauging interface and the detector bookkeeping of a code switch rather than the layer-code geometry. Its entry and exit satisfy the binary stabilizer transition identities. Noisy repeated extraction produces one detector history, decoded by minimum-weight perfect matching using Stim's graph-like decomposition of the circuit detector error model. The headline circuits admit decomposition without ignored failures; this does not make a matching decoder exact for correlated depolarizing faults. Physical shots are sampled from the full circuit noise model.

The SI1000-like model uses two-qubit depolarizing strength p, single-qubit and ordinary idle strength $p/10$, measurement flips $5p$, reset noise $2p$, and data idle-wait noise $2p$. The archived campaign comprises 255 parameter points and 4.534×10^6 shots for sizes $L = 3, 5, 7, 9, 11$ and gauged durations $G = d, 2d, 3d$, where $d = L$ is the code distance. All 15 noiseless controls have zero failures.

For every duration, the failure rate decreases monotonically with L across all five sizes at every tested $p \le 0.004$ (Fig. 4). Consecutive-size Wilson intervals are separated at every tested p from 0.0025 to 0.00375 for all three durations. At $p = 0.0035$, increasing L from 3 to 11 reduces the failure rate from 0.1111 to 0.0324 for $G = d$, and from 0.1326 to 0.0428 for $G = 3d$. At $p = 0.0015$ and $G = d$, the reduction is approximately 6.5×10^2.

The smallest-versus-largest crossing medians are 4.471, 4.431, and 4.328×10^{-3} for $G = d, 2d, 3d$, with conditional 95% posterior bands $[4.426, 4.522]$, $[4.392, 4.466]$, and $[4.265, 4.393] \times 10^{-3}$. Tripling the gauged duration shifts the median by about 3.2%. These are finite-size diagnostics of a surface-code gauging model; they do not estimate the architecture theorem's asymptotic threshold, which concerns layer codes and a committed minimum-weight decoder.

A complete joint measurement on explicit layer-code blocks tests what the surface-code campaigns cannot: the reported bit, both logical sectors, spectator logicals, and an entangled input. Two $[[4, 2, 2]]$-input blocks ($n = 224$, two logical qubits each, distances 5 and 4) measure $\overline{X}_1^{(a)}\overline{X}_1^{(b)}$ by gauging on a planar grid attached to their single-layer slabs, with ancilla-based extraction of every check and uniform circuit noise. Each logical qubit starts maximally entangled with a noiseless reference, and eight observables certify the output Choi state. Without noise every detector and observable is deterministic, so the circuit implements the projective measurement with the correct outcome and leaves the spectators untouched. We then minimize the number of faults that flips each observable undetectably, reporting exact values or certified brackets according to the solver status. With a bridge of $w = 1, 2, 3$ grid rows between the blocks, the fault distance of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ is exactly w; for $w = 1$ a single fault suffices. This is the cut obstruction of Sec. III B at circuit level. A grid spanning both blocks restores it to exactly 5, the blocks' X distance, The reported-bit distance is 4 and the tested spectator X-frame distance is 4 for all five interfaces. Several other distances remain bracketed; Table IV lists those bounds explicitly.

A second campaign repeats the measurement on blocks of spacing $s = 2$ to 5 (block distances up to $d_X = 11$, up to 1400 qubits per block), with $2s + 1$ rounds per phase and 3.0×10^6 shots, decoded in each CSS sector separately (Sec. VIII C); on identical shots the full-model BP+LSD decoder has a lower estimated failure rate through the grid (about half in the paired X test) and a similar estimate through the bridge at the tested points. This comparison is restricted to those paired points. At $p = 10^{-3}$ the failure rate of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ through the one-row bridge rises from 3.9% at $s = 2$ to 8.8% at $s = 5$, because the undetectable bridge fault has more rounds in which to occur, while through the spanning grid it falls from 8.2×10^{-4} to about 10^{-4} and is statistically compatible with the estimates for the same blocks left idle. This does not establish equivalence or a resolved improvement at every successive spacing. For the spanning grid each unit of spacing suppresses failures by a factor 4.8 in the X experiment and 2.6 in the Z experiment at $p = 10^{-3}$, falling

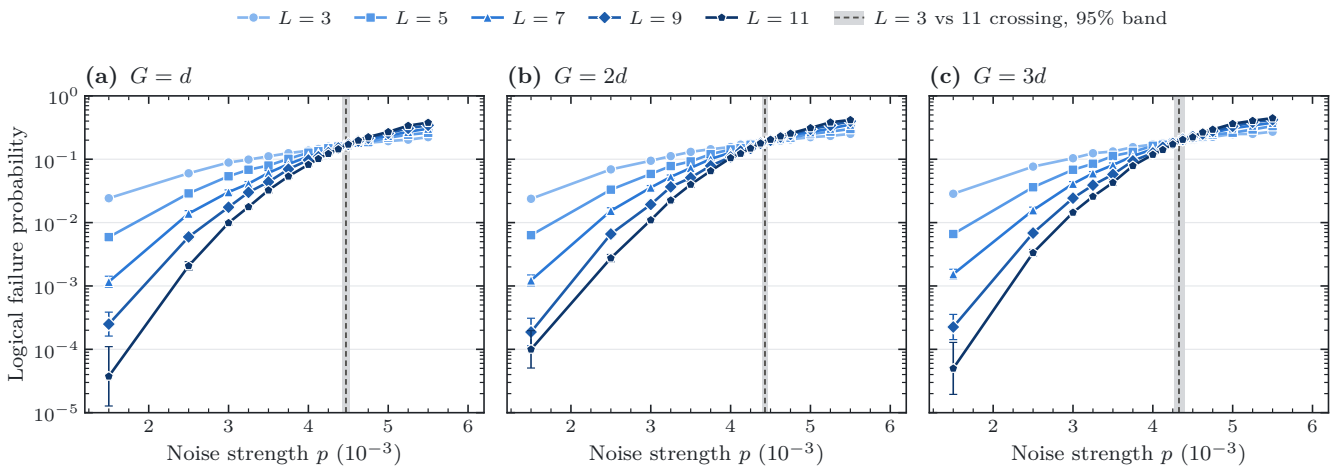


FIG. 4. Finite algebraic code-switch diagnostics on the planar surface code of distance L, redrawn from the archived shot counts. Panels (a)–(c) use gauged durations $G = d, 2d, 3d$, respectively. Markers identify the five sizes; bars are pointwise Wilson 95% intervals. Vertical dashed lines and shaded bands show the $L = 3$ versus $L = 11$ crossing medians and conditional 95% Jeffreys posterior intervals. The horizontal axis is expressed in units of 10^{-3}. These curves characterize the implemented algebraic transition model; the bands show posterior uncertainty under the stated interpolation rule and exclude systematic finite-size uncertainty.

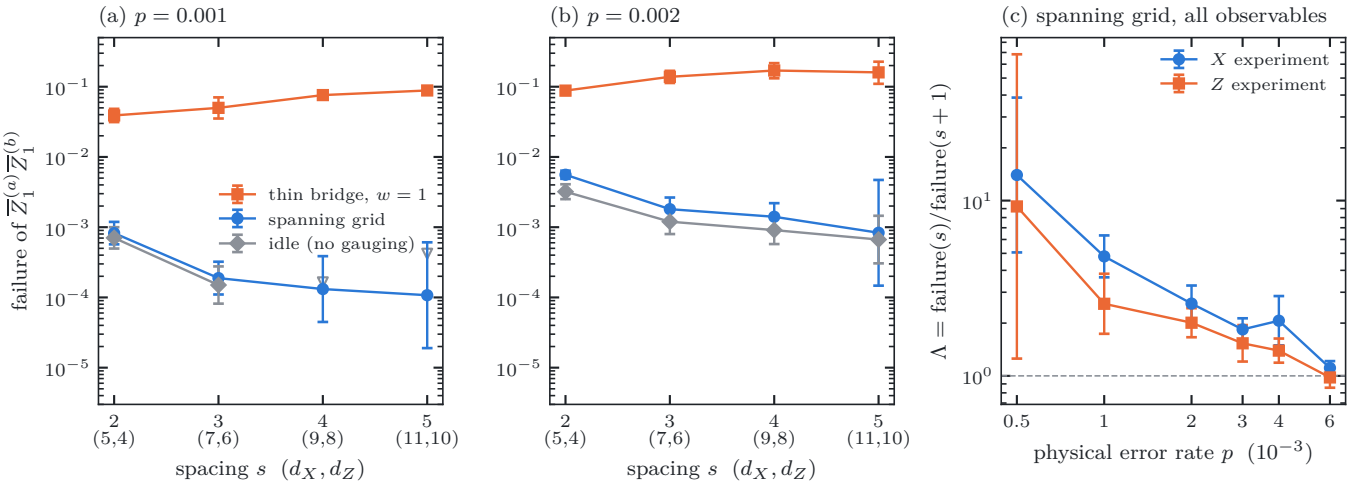


FIG. 5. Joint measurement at growing block size. (a,b) Failure of the measured pair's Z sector through the one-row bridge, the spanning grid, and for the same blocks idling without gauging, at $p = 10^{-3}$ and 2×10^{-3}; bars are Wilson 95% intervals and open triangles the upper end at zero failures. At low noise the estimates show worsening bridge performance and overall grid suppression compatible with idle memory; adjacent sizes need not have statistically resolved differences. (c) Suppression factor per unit of spacing for the spanning grid (all observables of each experiment), weighted fit over the available sizes with 95% intervals; it reaches one near $p \approx 6 \times 10^{-3}$.

to one near $p \approx 6 \times 10^{-3}$ (Fig. 5; raw counts in Table I). At $s = 3$ the fault distance of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ through a bridge of $w = 1, 2, 3$ rows is still exactly w, although the blocks' X distance is now 7.

A third campaign replaces the colocated ancillas by the routed nearest-neighbor schedule of Sec. VIII C, with SWAP-shuttled ancillas, colored conflict classes, and idle noise in every time step, and grows the input at $s = 2$: $[[4, 2, 2]]$, a six-qubit band code, and the Steane code, giving blocks of 224, 720 and 1792 qubits and circuits of up to 16,293 qubits. Without noise all 28 measurement circuits implement the measurement exactly, and the 12 idle controls also give silent detectors and correct observables. Through the one-row bridge the fault distance of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ is exactly 1 for all three inputs, although the blocks' X distance grows from 5 to at least 18. On the $[[4, 2, 2]]$ blocks with $R = 5$ rounds per phase the spanning grid gives exactly 5, the same as the blocks left idle; through both interfaces the reported bit has exactly $5 = R$, and the spectator X frames have exactly $4 = d_Z$, as in idle

TABLE I. Selected raw counts for the scaling experiment at $p = 10^{-3}$: failures/shots of the measured pair's joint Z frame. Each block uses a $[[4, 2, 2]]$ input, with $(d_X, d_Z) = (2s + 1, 2s)$ and $R = 2s + 1$ rounds per phase. Idle controls run for the same $3R$ rounds without gauging. Decoding is per-sector BP+LSD; $0/n$ has Wilson 95% upper end about $3.8/n$.

Interface	$s = 2$: $(5, 4)$	$s = 3$: $(7, 6)$	$s = 4$: $(9, 8)$	$s = 5$: $(11, 10)$
One-row bridge	78/2000	30/600	342/4500	252/2850
Spanning grid	28/34000	13/69000	3/22800	1/9300
Idle control	31/44000	10/66600	0/23400	0/9150

memory. Where both are resolved, routing leaves the distances of the colocated circuit unchanged. On the band blocks the reported bit has an undetectable pattern of weight 6 at $R = 7$. The block itself explains this weight: the lightest Z-type operator that anticommutes with the measured logical has weight 6, and the archived circuit witness realizes weight six. Together with the repeated-record witness, these bound the reported-bit fault distance above by $\min\{R, 6\}$ in this noise and measurement model; a smaller routed-circuit minimum remains unresolved. At the tested noise strengths the sampled failure rates of these long routed circuits do not distinguish the interfaces, so we report exact fault distances where certified and explicit bounds otherwise.

The deterministic checks include the complete-code slab certificate and negative controls in Sec. VIII B. These support the finite construction and remain separate from the arbitrary-size proofs.

VII. DISCUSSION AND CONCLUSIONS

The slab theorem gives a local route from layer-code storage to logical access. Prefix-parity repair exposes logical operators in single-layer slabs of an explicit realization, and amplified grid gauging preserves static distance. The local primitives support universal logical operations and deterministic resource distillation. Under the uniform scheduled-circuit hypotheses, the architecture bound separates an extensive Wf space cost from the minimum $f^{3/2}$ cost of one encoded cell in bounded-aspect logical boxes. Establishing these hypotheses for the full adaptive implementation is a remaining proof requirement; the finite certificates verify only their tested instances.

The next steps are a uniform proof of the scheduled detector, transfer, and decoder-witness conditions; a practical decoder for deformation histories, with changing checks, inherited frames and records, and joint measurements (the layer-code decoders of Ref. [30] are a natural starting point, although their memory guarantees do not yet cover code switching); routed circuit-level benchmarks on inputs that grow into the balanced regime, with a shorter routed schedule and a decoder close to minimum weight; bounds on how many logical measurements can share the slab interfaces, compared with parallel code surgery [3] under three-dimensional locality; and factories whose constants make the space–time estimate competitive, possibly in a different code with a protected local transfer. Random layer codes built from random CSS inputs [30] are a natural candidate for reducing those constants; the explicit realization applies to any CSS input, but the resource accounting would need their parameter scaling. The conditional architecture theorem uses minimum-weight committed decoding and local-stochastic Pauli noise. Extensions to coherent or leakage noise require an additional fault-path analysis; deterministic selection of the input component codes is a separate constructive problem. Comparing the space bound with an extensive storage lower bound requires the restricted regime $W = \Omega(\sqrt{f})$, so that the one-cell term is not dominant.

VIII. CONSTRUCTION AND COMPUTATIONAL METHODS

A. Lattice specification

Let $H_X \in \mathbb{F}_2^{m_X \times n_{\text{in}}}$ and $H_Z \in \mathbb{F}_2^{m_Z \times n_{\text{in}}}$ be the check matrices of an input CSS code with $H_X H_Z^T = 0$, every check of weight at least two, and $K = n_{\text{in}} - \operatorname{rank} H_X - \operatorname{rank} H_Z$. Fix an integer spacing $s \geq 2$ and set

$$\begin{aligned} Y_q &= sq, \quad X_j = s(j + 1), \quad Z_i = s(i + 1), \\ L_x &= s(m_Z + 1), \quad L_z = s(m_X + 1), \end{aligned} \tag{26}$$

with $q = 1, \ldots, n_{\text{in}}$, $j = 0, \ldots, m_Z - 1$ and $i = 0, \ldots, m_X - 1$. For a check c write $q_1^{(c)} < \cdots < q_{k_c}^{(c)}$ for its support in increasing order; $q_1^{(c)}$ is its *first*, $q_{k_c}^{(c)}$ its *last*, and the others its *middle* incidences.

a. Sheets. Every sheet is an edge surface code on a rectangle of the cubic lattice: one qubit per unit edge of the sheet, an X check (*star*) on each vertex acting on the sheet edges at that vertex, and a Z check (*face*) on each unit square acting on its boundary edges.

- Data sheet D_q ($q = 1, \ldots, n_{\text{in}}$): the plane $y = Y_q$ with $0 \leq x \leq L_x$, $0 \leq z \leq L_z$. The rows $z = 0$ and $z = L_z$ carry no x-edges and no stars.
- Z-check sheet C_j: the plane $x = X_j$ with $Y_{q_1} \leq y \leq Y_{q_k}$ ($q_t = q_t^{(Z_j)}$) and $0 \leq z \leq L_z$. The rows $z = 0, L_z$ carry no y-edges and no stars.
- X-check sheet A_i: the plane $z = Z_i$ with $Y_{q_1} \leq y \leq Y_{q_k}$ ($q_t = q_t^{(X_i)}$) and $0 \leq x \leq L_x$; every vertex carries a star.

Distinct sheets own distinct qubits, also on coincident lattice edges, with one exception: the first and last z-edge columns of C_j *are* the z-lines $(X_j, Y_{q_1}, \cdot)$ and $(X_j, Y_{q_k}, \cdot)$ of D_{q_1} and D_{q_k}, and the stars of C_j on those columns are merged into the corresponding data stars. A sheet pair without an input incidence therefore crosses transparently. At most two qubits share a lattice edge.

b. Line junctions. Write $d_\mu(v)$, $c_\mu(v)$, $a_\mu(v)$ for the μ-directed edge $v \to v+\hat{\mu}$ of D, C, A respectively. The input incidences act by adding one qubit to existing checks:

(Z) for $q = q_t^{(Z_j)}$ and every vertex $u = (X_j, Y_q, z)$ of the z-line: for $1 \le z < L_z$, the final support of the D_q star at u contains $c_y(u)$ if $t < k$ and $c_y(u - \hat{y})$ if $t = k$; for $0 \le z < L_z$, if q is a middle incidence, the C_j face on $[Y_q, Y_q + 1] \times [z, z + 1]$ gains $d_z(u)$. (For the first and last incidence the adjacent C_j face already contains d_z through the shared column. The endpoint star merge already includes the indicated c_y edge; do not add it a second time modulo two.)

(X) for $q \in \operatorname{supp} X_i$ and every vertex $u = (x, Y_q, Z_i)$ of the x-line: the A_i star at u gains $d_z(u)$, and the D_q face on $[x, x + 1] \times [Z_i, Z_i + 1]$ gains $a_x(u)$ for $0 \le x < L_x$.

c. Walls on crossing lines. A Z-check sheet C_j and an X-check sheet A_i cross along the y-line $\ell_{ij} = \{(X_j, y, Z_i)\}$ for y in the overlap $[\ell, h]$ of their y-ranges. On this line we allow two further one-qubit additions, with binary coefficients σ_y $(\ell \le y \le h)$ and τ_y $(\ell \le y < h)$: the A_i star at (X_j, y, Z_i) gains $\sigma_y\, c_z(X_j, y, Z_i)$, and the C_j face on $[y, y + 1] \times [Z_i, Z_i + 1]$ gains $\tau_y\, a_y(X_j, y, Z_i)$. The coefficients are the unique solution of the commutation conditions of the checks: the additions change the overlap of the A star at y' and the C face on $[y, y + 1]$, $y' \in \{y, y + 1\}$, by $\sigma_{y'} + \tau_y$. All commutation equations, including those with checks immediately outside the overlap and adjacent data faces, are imposed. If the C face on $[\ell - 1, \ell]$ exists it fixes σ_ℓ; if the overlap starts at the shared first C column, the adjacent D faces force $\sigma_\ell = 0$. The right endpoint is handled in the same way. Empty overlap requires no wall; a single-vertex overlap has only its one σ and the endpoint equations, with no τ. The resulting path has fixed ends; summing its equations gives $|\operatorname{supp} X_i \cap \operatorname{supp} Z_j|$ modulo two. The system is therefore solvable, uniquely, exactly when that overlap is even, which holds for every CSS input. This completes the check-matrix specification.

B. Complete-code verification and controls

All finite checks use exact GF(2) linear algebra on the complete check matrices; no boundary check is omitted. For seven input codes and spacings $s = 2, 3$ where feasible, the tests verify commutation, solvability of every wall system, the Euler count $n - \#X - \#Z = n_{\text{in}} - m_X - m_Z$, the dimensions of the relation spaces, and $k = K$. Table II records the resulting logical count and slab-pairing rank.

TABLE II. Complete-code verification. All rows have zero anticommuting check pairs. The encoded dimension and slab-pairing rank both equal K; the last column is the incorrect logical count of the single-chart control.

Input	s	qubits n	k = pairing rank	control k
$[[4, 2, 2]]$	2	224	2	4
$[[4, 2, 2]]$	3	504	2	4
$[[6, 4, 2]]$	2	352	4	6
$[[6, 4, 2]]$	3	792	4	6
Steane $[[7, 1, 3]]$	2	1792	1	10
Steane $[[7, 1, 3]]$	3	4032	1	10
Hamming $[[15, 7, 3]]$	2	6600	7	29
Surface $[[13, 1, 3]]$	2	10314	1	15
Random HGP 1	2	10936	2	20
Random HGP 2	2	10512	2	18

For each stabilizer, the overlap with the symbolic prefix operator is compared with the first/middle/last forms of Eqs. (2) and (3), at every height and every dual plane. For a single check of weight $2 \le k \le 6$, all 2^{2k-1} incidence/backbone assignments are enumerated. Precisely the 2^{k-1} even incidence words with their prefix backbones have zero syndrome; odd words and single-segment flips fail. Every basis input is checked for stabilizer commutation, mapping of input stabilizers to stabilizers, and preservation of the logical pairing. Here HGP denotes a hypergraph-product code. The single-chart negative control identifies every incidence column of a check sheet with the corresponding data column, including middle incidences, thereby removing the independent check-sheet pass-through channel. Its precise construction is included in the source archive. For this control, its encoded dimension differs from K for every tested input. Walls with odd check overlap are inconsistent; the even-overlap controls are solvable.

C. Simulation and inference protocol

a. Layer-code circuits. The initial two-block experiment has 12 port qubits per block. Every check is measured with its own ancilla and CNOTs, X checks before Z checks, for four rounds before, during, and after gauging. Grid edges are prepared in $|0\rangle$ and read in Z; edge readouts update the Pauli frame. Uniform circuit noise applies two-qubit depolarizing noise of strength p after each CNOT, one-qubit depolarizing noise on data and edge qubits each round, and flips of probability p after resets and before measurements, including final readout. This differs from the SI1000-like surface-code model specified in Sec. VI.

Every input logical qubit is entangled with a noiseless reference. An X-basis experiment tests the reported parity and four logical X frames; a Z-basis experiment tests the measured pair's joint Z frame and two spectator Z frames. These eight observables certify the output Choi state in the noiseless circuit. CP-SAT minimizes the

TABLE III. Initial two-block experiment, $s = 2$, four rounds per phase, $p = 0.002$. The target is the measured pair's joint Z frame. Each rate uses 1000 shots; brackets are Wilson 95% intervals. A distance bracket records a proven lower bound and an undetectable witness.

Interface	fault dist.	failures	rate (%) [95% interval]
Bridge $w = 1$	1	71/1000	7.1 [5.7, 8.9]
Bridge $w = 2$	2	68/1000	6.8 [5.4, 8.5]
Bridge $w = 3$	3	5/1000	0.5 [0.2, 1.2]
Spanning grid	5	0/1000	0.0 [0.0, 0.38]
Doubled grid	4–5	2/1000	0.2 [0.05, 0.73]

number of undecomposed detector-error mechanisms that flip a chosen observable with zero detector syndrome. A distance is called exact only when optimality is proved; otherwise the proven lower bound and a found witness give a bracket. The distance computation uses undecomposed detector signatures at unit cost, merging identical signatures. In the stated Pauli noise model, allowed errors at one location are closed under multiplication (up to phase), so multiple selected labels at one location can be combined into one allowed error with the same net signature and no greater cost. Hence the minimum equals a physical-location minimum for this model. A graphlike decomposition into multiple edges is not assigned multiple physical-fault costs. The disjoint-error probability approximation in the DEM conversion affects decoder weights, not this support-count minimum.

Here the doubled grid replaces each planar-grid edge by two parallel ancillas. Its cycle checks include the original face cycles on one copy and a weight-two check joining the two copies of each edge. It is the multiplicity-two version of the construction in Appendix B, not an assertion that two copies achieve the worst-case three-dimensional amplification constant.

b. Decoding and sampling. The initial experiment uses full-model belief propagation with localized statistics decoding (BP+LSD): minimum-sum BP, 30 iterations, scaling factor 0.9, serial schedule, and LSD order 4. Four batches of 250 shots per basis use distinct recorded integer seeds. The larger campaign keeps these decoder settings but decodes each CSS sector separately; all physical faults are still sampled, while inter-sector correlations are ignored by the decoder. It compares spacings $s = 2, 3, 4, 5$ with $R = 2s + 1$ rounds per phase, and the same blocks idling for $3R$ rounds. Noise strengths are $p \in \{0.5, 1, 2, 3, 4, 6\} \times 10^{-3}$.

The runner stops scheduling new work for a parameter point once its aggregate any-observable failure count reaches 200, its nominal shot cap is reached, or the shared 36-hour scheduling budget expires. Already assigned worker sessions can finish several batches and overshoot these targets; the data are aggregated by unique task and batch identifiers. Batch sizes for $s = 2, 3, 4, 5$ are respectively $2000, 600, 300, 150$, and the nominal shot caps are $2 \times 10^6, 600000, 200000, 80000$. The scheduler allows in-flight work after the budget, with a hard stop one hour later. Thus 200 is a scheduling target, not a cap on the reported counts. Ten of 206 points have no data because of memory limits; these near/above-crossing points are not used for the reported below-crossing comparison. Table I gives actual counts at $p = 10^{-3}$. Zero observed failures give an upper confidence bound, not proof of zero failure probability. Paired tests on identical shots compare the full-model and sector decoders; the latter has a higher estimated grid failure rate at the checked points. Fits and error bars characterize this heuristic decoder on finite circuits, not the theorem's minimum-weight decoder.

c. Routed extraction. Each ancilla follows a depth-first tour of a reserved nearest-neighbor spanning tree, applies a data–ancilla CNOT on the first visit to each support qubit, retraces its path, and is read out. A greedy coloring separates overlapping extraction words. Every CNOT and SWAP has two-qubit depolarizing noise p, preparation and readout flip with probability p, and data and active routing qubits incur idle noise p at each waiting time step. The routed results in Sec. VI are exact distances or explicit bounds; sampled failure rates are not used to claim interface separation for these long circuits.

d. Statistical interpretation. The plotted Wilson 95% intervals are approximate pointwise binomial intervals. Their nominal fixed-shot coverage is not guaranteed under the implemented failure-dependent stopping rule; they are descriptive here. For the surface-code crossings, independent Jeffreys posteriors $\mathrm{Beta}(k+1/2, n-k+1/2)$ are sampled at each noise point for $L = 3$ and $L = 11$. The first sign change in their rate difference is interpolated linearly in p. Draws without an interior crossing are excluded. The reported medians and central intervals are conditional on that event and omit systematic finite-size and model uncertainty.

For Fig. 5(c), let k_s count shots failing any observable in the chosen basis. The archived fit uses $\log(k_s/n_s) = a - bs$, weights k_s, and only sizes with $k_s > 0$. It reports $\Lambda = e^b$ and $\exp(b \pm 1.96/\sqrt{\sum_s k_s(s - \bar{s})^2})$ with weighted $\bar{s}$. These are approximate fit intervals based on a rare-event variance model. Omitting zero counts and using adaptive sample sizes can bias the fit; these suppression factors are descriptive diagnostics. Appendix F provides the counts needed to reproduce the fits, which differ from the single-observable counts in Table I.

DATA AND CODE AVAILABILITY

The accompanying versioned archive `IRIS_FINAL_SOURCE.zip` contains the editable manuscript, the supplied input constructions, simulation and analysis scripts, archived per-batch seeds and counts, decoder environment records, and available solver outputs. Its manifest records SHA-256 digests. The revision uses the supplied data without adding new Monte Carlo shots. Appendix F identifies the files underlying the displayed numbers. The archive is a local deliverable, not a claim of public deposition or independent reproduction of every campaign. Some retained author records contain earlier manuscript

TABLE IV. All tested observable types from the initial two-block distance file, $s = 2$ and $R = 4$. A single integer has solver status OPTIMAL; $[l, u]$ gives its proven lower bound and best witness when optimality is unresolved. The spectator Z distances are not all resolved. $X_1^{(a)}$ is a retained measured-pair frame, and $X_2^{(a)}, Z_2^{(a)}$ are spectator frames.

Interface	reported bit	$X_1^{(a)}$	$X_2^{(a)}$	$Z_1^{(a)}$	$Z_1^{(b)}$	$Z_2^{(a)}$
Bridge $w = 1$	4	[4,10]	4	1	[4,12]	
Bridge $w = 2$	4	[4,12]	4	2	[5,12]	
Bridge $w = 3$	4	[4,10]	4	3	[5,13]	
Spanning grid	4	[4,10]	4	5	[5,12]	
Doubled grid	4	[4,10]	4	[4,5]	[5,12]	

text; the revised manuscript and its explicit hypotheses define the claims of this edition.

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Appendix A: Logical-space and distance proofs

All vector spaces in these proofs are over $\mathbb{F}_2$. Logical Pauli classes are taken modulo stabilizers and irrelevant phases. Let g_X, g_Z denote the numbers of listed lattice stars and faces, and r_X^{rel}, r_Z^{rel} the dimensions of their relation spaces. These relation dimensions are distinct from the numbers of disjoint ports in the main text.

1. Commutation and relation maps

The junction rules in Sec. VIII A preserve commutation except at a crossing of an A_i sheet and a C_j sheet. For example, a data star and an adjacent C face share both the data z-edge and the added c_y edge. An A star and an adjacent data face share both the added d_z and an a_x edge. At an A_i–C_j crossing, each common input qubit contributes one unmatched overlap. The wall equations add $\sigma_{y'} + \tau_y$ to the overlap of the star at y' and the face on $[y, y+1]$, for $y' = y, y+1$. Successive equations determine the variables along a path, with the endpoint equations specified in Sec. VIII A. Their sum is the parity of $\operatorname{supp} X_i \cap \operatorname{supp} Z_j$. This is zero by $H_X H_Z^T = 0$, so the endpoint conditions agree and the solution is unique. An empty overlap has no variables; for a one-vertex overlap only the two endpoint conditions on the same variable remain. Thus all checks commute.

The sheet counts, assigning shared boundary columns to data sheets, are

$$\begin{aligned} n_D &= L_x(L_z-1) + (L_x+1)L_z, \\ g_{X,D} &= (L_x+1)(L_z-1), \quad g_{Z,D} = L_x L_z, \\ n_C &= w(L_z-1) + (w-1)L_z, \\ g_{X,C} &= (w-1)(L_z-1), \quad g_{Z,C} = wL_z, \\ n_A &= L_x(w+1) + (L_x+1)w, \\ g_{X,A} &= (L_x+1)(w+1), \quad g_{Z,A} = L_x w. \end{aligned} \tag{A1}$$

Here w is the sheet's width in lattice edges. Data sheets contribute $+1$ and either check sheet contributes -1 to the Euler count, so

$$n - g_X - g_Z = n_{\mathrm{in}} - m_X - m_Z. \tag{A2}$$

The junction and wall additions change supports without adding generators or qubits. Redundant input rows cannot be discarded in this count.

For an X relation, assign a binary selection coefficient to every star. Each edge owned by A_i belongs to its two endpoint A_i stars and no other star, so the selection is constant, say α_i, on that sheet. On the data column $x = 0$, a data z-edge belongs to the data stars at its endpoints when present, and to an A_i star at a corresponding incidence. Summing the vanishing edge coefficients along the column, where there are no stars at $z = 0, L_z$, gives

$$\sum_{i:q\in X_i} \alpha_i = 0 \quad \text{for every } q, \qquad \alpha^T H_X = 0. \tag{A3}$$

The coefficients on the data stars are determined from the bottom boundary; data x-edges make them constant

along each row. Coefficients on the remaining C stars are then determined from their bottom boundary, since $c_z(y,0)$ meets just one such star. Thus the map from a lattice X relation to α is injective.

For a Z relation, the edges owned by C_j imply a constant face coefficient γ_j on that sheet. Data x-edges imply that the data face coefficient is independent of z; write it as $b_q(x)$. The data z-edges give

$$\begin{aligned} b_q(0) = b_q(L_x - 1) = 0, \\ b_q(x-1) + b_q(x) = \sum_{j:X_j=x,\ q\in Z_j} \gamma_j. \end{aligned} \tag{A4}$$

Summing yields $\gamma^T H_Z = 0$. The data coefficients are determined by γ, and the A-face coefficients are determined from $x = 0$, where each a_y edge meets one face. Hence this relation map is also injective. Consequently,

$$\begin{aligned} k &= n - g_X - g_Z + r_X^{\mathrm{rel}} + r_Z^{\mathrm{rel}} \\ &\le n_{\mathrm{in}} - m_X - m_Z + (m_X - \operatorname{rank} H_X) \\ &\quad + (m_Z - \operatorname{rank} H_Z) \\ &= K. \end{aligned} \tag{A5}$$

2. Slab pairing, stabilizers, and translation

The X slab for $x \in \ker H_Z$ contains all data z-edges at height z_0 on selected data sheets. On C_j, the segment after incidence t has coefficient $p_t = \sum_{h\le t} x_{q_h}$. More precisely it contains the C_j-owned z-edges with $Y_{q_t} < y \le Y_{q_{t+1}}$, excluding the shared last column. Local overlaps with faces are the endpoint and middle relations of Eqs. (2) and (3). They all vanish. Walls add only a_y edges to these faces and hence do not change this calculation.

For $z \in \ker H_X$, the dual slab at x_0 contains every data z-edge at that x coordinate on selected data sheets and the A_i-owned y-edges with $Y_{q_t} \le y < Y_{q_{t+1}}$, weighted by the analogous prefix parity. It commutes with every star. Opposite slabs intersect on exactly one data edge for each input coordinate selected by both; their other supports are sheet owned and disjoint. Their pairing is therefore

$$\langle \widetilde{X}(x; z_0), \widetilde{Z}(z; x_0) \rangle = x \cdot z. \tag{A6}$$

Choose input logical bases with pairing the identity. Their lifts give K independent logical pairs and thus $k \ge K$. Together with Eq. (A5), this proves $k = K$.

This also closes the two identifications needed in the main text without assuming them in the rank argument. If x is an input X stabilizer, its lift pairs trivially with every lifted Z logical. Those lifted logicals span the full opposite logical space because $k = K$, so the lift is a lattice stabilizer. The dual statement follows identically. The difference of two heights of an X lift has zero pairing with every Z lift by Eq. (A6), so it too is a stabilizer. The same holds for translating Z planes. Hence the lifts induce an isomorphism of input and lattice logical spaces at every allowed position.

3. The per-plane projection

Fix a nontrivial lattice X logical O and a plane $x = x_0$. Decompose the dual slab into disjoint pieces: one vertical data column per input qubit and one A_i backbone segment per interval between successive incidences. Let a_q and $b_{i,t}$ be the parities of the intersection of O with these pieces. Define the input vector

$$\pi_{x_0}(O) = \sum_q a_q e_q + \sum_{i,t} b_{i,t} \sum_{h\le t} e_{q_h^{(X_i)}}. \tag{A7}$$

For every $z \in \ker H_X$, its dot product with z is the commutator of O with the Z lift of z in this plane. Taking z to be an input Z stabilizer proves $\pi_{x_0}(O) \in \ker H_Z$. Taking a logical z conjugate to O proves that the projected class is nontrivial. This argument holds in every plane because the lifts represent the same class.

Replacing a prefix by its complementary suffix adds the input X_i stabilizer. Choose the shorter representative of each prefix; it has weight at most $\lfloor w_X/2 \rfloor$. Each parity-one piece contains at least one occupied qubit of O, and distinct pieces in a plane do not share qubits. Thus a representative of $\pi_{x_0}(O)$ has weight at most η_X times the number of occupied qubits of O in the plane. Its weight is at least d_X^{in}. Summing over the $L_x + 1$ disjoint planes proves Eq. (4). Exchanging the two constructions and summing over the L_z disjoint layers proves Eq. (5). Complete slabs have $O(N^2)$ qubits for balanced bounded-degree inputs: there are $O(N)$ data pieces of length $O(N)$ and $O(N)$ bounded-weight checks with total backbone length $O(N^2)$. This supplies the matching upper bound for every pure logical class and, by taking a product of two slabs, for every mixed Pauli class.

Appendix B: Grid expansion and unequal-distance protection

1. A discrete relative-expansion bound

Let P_a be the path with a vertices. Side lengths in this appendix count vertices, not edges. Let $G_0 = P_a \square P_b$ with $2 \le a \le b$, and let $h(G_0)$ be its edge Cheeger constant. We first prove

$$h(G_0) \ge 1/b. \tag{B1}$$

For $|S| \le ab/2$, regard the a rows as having length b. If there is both a full and an empty row, every column has a vertical boundary edge, so $|\delta S| \ge b \ge |S|/b$. If no row is full, every nonempty row has a horizontal boundary edge, giving $|\delta S| \ge |S|/b$. If a full row exists but no row is empty, all columns meet S and at most $|S|/a \le b/2$ columns are full. At least $b/2$ columns therefore have a vertical boundary edge, again giving $|\delta S| \ge |S|/b$. Complementation handles larger sets.

Now take $G = G_0 \square P_\ell$ and let U occupy at most t of its parallel G_0 layers, with $|U| \ge 2$. For any cut set S,

denote its layer slices by S_j. Fix one layer j_0, and put

$$\mu = \min\{|S_{j_0}|, |V(G_0) \setminus S_{j_0}|\}, \qquad m = \max_j |S_j \triangle S_{j_0}|. \tag{B2}$$

The vertical boundary has at least m edges: along a path of layers to a maximizing slice, every changed vertex crosses a boundary. Each horizontal slice has minority size at least $\max\{0, \mu - m\}$. Writing $c = \min\{1, \ell h(G_0)\}$, the total boundary obeys

$$|\delta S| \geq m + c\max\{0, \mu - m\} \geq c\max\{\mu, m\}. \tag{B3}$$

Choose the minority side of the reference slice; in each port layer it contains at most $\mu + m$ vertices. Hence

$$\min\{|S \cap U|, |U \setminus S|\} \leq t(\mu + m) \leq 2t\max\{\mu, m\}. \tag{B4}$$

If $\max(a, b, \ell)/\min(a, b, \ell) \leq A$, then $c \geq 1/A$ and $\beta(G, U) \geq 1/(2At)$, proving Eq. (9). Any coordinate orientation can be used. Ports need not fill their layers. Constant lattice refinement separates the finitely many local port labels and changes A, t only by constants. Cases with fewer than two ports are excluded from this relative-expansion definition and treated directly.

2. Parallel edges and the cycle space

Replace every edge e by m_0 parallel copies and attach all copies to the same vertex stars. Use one designated copy for each original pairing path and each elementary square cycle. In addition, measure $Z_{e,1}Z_{e,r}$ for $2 \leq r \leq m_0$. These checks span the entire cycle space of the multigraph: a cycle can first be moved to designated copies using the two-edge cycles and then reduced using the rectangular grid's square cycles. Thus no additional cycle logicals are introduced. Every cut contains all m_0 copies of each original cut edge, so

$$\beta(G^{(m_0)}, U) = m_0\beta(G, U). \tag{B5}$$

Taking $m_0 = \lceil 2At \rceil$ makes this at least one. At fixed m_0, star weights, cycle weights, and check incidences remain bounded. The copies can be placed in a fixed refinement cell near the original edge; bounded congestion then permits a constant-depth local extraction schedule for each static check catalog. Uniform properties across an entire adaptive history still require Assumption IV.1.

3. The protected quotient for unequal blocks

Here use the dual convention and gauge $M = \overline{Z}_1\overline{Z}_2 = Z(D_1)Z(D_2)$ on $C_1 \otimes C_2$. Let C_2 have distance at least d_2. Include all graph cycle checks. Let $\mathcal{Q}_1$ be generated by deformed images of the logical operators on C_1 that commute with $\overline{Z}_1$. An X image is completed by edges with boundary equal to its port incidence; a Z image needs no completion. Different completions differ by a cycle stabilizer. Because M is measured, $\overline{Z}_2$ and $\overline{Z}_1$ represent the same class in $\mathcal{Q}_1$.

Lemma B.1 (Static protected quotient)**.** *The minimum weight of a deformed logical class outside $\mathcal{Q}_1$ is at least* $\min\{1, \beta(G, U)\}d_2$.

Proof. For an X normalizer, restriction to the original blocks is in their original X normalizers; the edge boundary equation also requires commutation with M. If its C_2 restriction is a stabilizer, remove its deformed lift. The remaining C_1 logical commutes with $\overline{Z}_1$, and differences of edge completions are cycles, so the class lies in $\mathcal{Q}_1$. Otherwise its C_2 data restriction already has weight at least d_2.

For a Z normalizer, the ancilla support is a cut δS, since it is orthogonal to every cycle. Multiplying by vertex stars on either side removes the cut and gives two original logicals differing by M. If the C_2 part of either belongs to $\langle S_Z(C_2), \overline{Z}_2\rangle$, the class is in $\mathcal{Q}_1$. Outside this subgroup both reductions have nontrivial C_2 parts, each of weight at least d_2. Put a equal to the original data weight, $m = \min\{|S \cap U|, |U \setminus S|\}$, and $b_0 = \min\{1, \beta\}$. The smaller-side reduction has weight at most $a + m$ and at least d_2, whereas

$$a + |\delta S| \geq a + b_0 m \geq b_0(a + m) \geq b_0 d_2. \tag{B6}$$

Finally, $\mathcal{Q}_1$ is generated by pure-type classes in this CSS setting. A mixed class outside it has a pure component outside it, of weight no larger than the mixed operator, giving the same bound. □

The lemma also permits C_1 to be a bare qubit, when its commuting logical subgroup is generated by Z_1. During teleportation the parity and final X_1 outcomes update only the traveling slot. This algebraic statement explains the exceptional resource degree of freedom. It does not by itself prove that every inherited syndrome at a noisy open boundary is harmless to spectators; that requirement is stated separately in Assumption IV.1.

Appendix C: A relative phenomenological slice bound

Consider a rectangle with at least one undeformed entry and exit round and R deformed rounds gauging $P = X(D)$. A fault is a single-qubit Pauli in a gap before a round or a flip of one check record. Edge preparation and readout flips are represented at their first and last gaps. Subtract the fault-free propagation of the same incoming (s, λ, ρ) and fix the same ideal random gauge outcomes. The relative input is zero; require zero relative syndrome and records at the right cut. The first and last cut records are included as boundary data, not silently replaced by perfect physical measurements.

Write E_t for the accumulated data-and-edge Pauli by round t. For a check c, with commutator $\sigma_c(E_t)$ and record fault $m_{c,t}$, its relative record is

$$\Delta r_{c,t} = \sigma_c(E_t) + m_{c,t}. \tag{C1}$$

Use consecutive-round detectors, preparation and readout relations, and the local entry/exit relations between a

Z check s and its lift $\widetilde{s} = sZ(\mu_s)$. The measured cycles generate the full cycle space. We prove Eq. (11) for a detector-silent pattern.

The CSS sectors can be treated separately; a Y mechanism contributes to both sectors but its count is not doubled when bounding either part. For the X sector, consecutive and entry/exit equations propagate zero relative Z-check records from the cuts through the rectangle. Let z_e be the parity of X faults on edge e, which is its relative final Z readout. Exit equations give

$$\sum_{e\in\mu_s} z_e = 0, \qquad \sum_{e\in c} z_e = 0 \quad \text{for every cycle } c. \tag{C2}$$

Thus $z = \delta S$. The final data error E_{d} commutes with the original Z checks by the output-syndrome condition. The single-slice operator $O = E_{\mathrm{d}}X(\delta S)$ commutes with the deformed checks. Multiplying its vertex stars over S leaves $L = E_{\mathrm{d}}X(f^{-1}(S\cap U))$, exactly the relative data frame after edge-readout correction. Complementing S changes L by P, which is trivial after the measurement. A nonzero surviving effect means that O is a nontrivial deformed logical, so $|O| \geq \widetilde{d}_X$. Every occupied data or edge timeline in O contains at least one original X-sector fault. Therefore this sector needs at least $\widetilde{d}_X$ faults for a nontrivial effect.

For the Z sector, original X-check records are chained from the zero input, and each vertex-star relative record is a constant b_u across the deformed rounds. The reported parity flip is $o = \sum_u b_u$. The output data error E_{d} is an original Z normalizer. If its class is nontrivial, its data support already costs at least d_Z faults. Otherwise a nonzero effect requires $o = 1$. If every deformed round contains an X-check record fault, there are at least R faults. If some round t^* has none, the data part $Z(B)$ of E_{t^*} commutes with every original X check. Each edge occurs in two vertex stars, whereas each port occurs in one, so

$$|B\cap D| = \sum_u \sigma_{A_u}(E_{t^*}) = \sum_u b_u = 1. \tag{C3}$$

Hence $Z(B)$ anticommutes with P and is a nontrivial original logical of weight at least d_Z. Its occupied data timelines cost at least that many faults. This sector therefore needs $\min\{d_Z, R\}$ faults. At least one sector has a nonzero effect, proving the claimed minimum. With no deformation, the same output argument gives the idle-memory distance $\min\{d_X, d_Z\}$. The X/Z-exchanged version applies to Z gauging. This is a lower bound for the stated rectangle and boundary conditions; exact attainment in each particular circuit requires its own witness or distance calculation.

Appendix D: Deterministic resource factories

1. Codes, orientations, and recovery

For $r = 3$ or 4, let $n_r = 2^r - 1$ and label coordinates by the distinct nonzero vectors $v_i \in \mathbb{F}_2^r$ in binary order. Let L_r be the row span of the $r \times n_r$ matrix with columns v_i, and define a CSS code with

$$S_X = L_r, \qquad S_Z = \langle L_r, \mathbf{1}\rangle^{\perp}. \tag{D1}$$

Here S_X, S_Z specify binary supports of stabilizer generators. The code encodes one qubit, with the fixed computational-basis convention

$$|b_L\rangle = 2^{-r/2}\sum_{x\in L_r} |x + b\mathbf{1}\rangle, \qquad b\in\{0,1\}. \tag{D2}$$

For $r = 3$ this is the Steane code; for $r = 4$ it is the punctured Reed–Muller code. All nonzero words of L_r have weight 2^{r-1}. Consequently $(S^\dagger)^{\otimes 7}$ has phase one on $|0_L\rangle$ and phase i on $|1_L\rangle$, while $(T^\dagger)^{\otimes 15}$ has phase one and $e^{i\pi/4}$ respectively. They implement logical S and T in this convention.

Prepare $|+_L\rangle$, apply the indicated transversal rotations by coordinatewise resource teleportation, measure the X-check syndrome, recover, and decode the logical output. The CSS encoder and decoder are fixed by Eq. (D2); Gaussian elimination realizes them with CNOTs, X/Z preparations, and X/Z measurements. They require no extra non-Clifford candidate slots. For the negative-angle rotations one may use $X|Y\rangle$ or $X|A\rangle$, equal up to phase to the corresponding conjugate resource. Each conditional $S^\dagger$ correction after a $T^\dagger$ injection is assigned one additional Y slot on the same coordinate.

The recovery table is completely specified by

$$R(0) = I, \qquad R(v_i) = Z_i. \tag{D3}$$

The columns v_i enumerate every nonzero syndrome once. Thus recovery is defined on every outcome and the factory always returns one output. In ideal operation all Z-check syndromes are zero; measuring them can also be included without changing this phase-error recovery rule.

2. Arbitrary defective candidates

Purify a candidate as $|0\rangle|\phi_0\rangle + |1\rangle|\phi_1\rangle$. In any fixed phase-teleportation branch, after the prescribed Pauli correction, the data-to-output map is diagonal in the computational basis. Relative to the ideal diagonal rotation it is therefore $I\otimes E_0 + Z\otimes E_1$. Tracing out the environment gives conditional Kraus operators in $\mathrm{span}\{I, Z\}$. This proof also covers mixed and correlated candidates by purification. A defective correction candidate remains diagonal on the same coordinate.

If at most one coordinate i is defective, the encoded block before recovery has error space $I\otimes E_0 + Z_i\otimes E_1$. The

two syndrome outcomes are 0 and v_i; Eq. (D3) corrects both. After correction the retained logical state is ideal and factors from the environment. This is a statement about correction of an error space, not about a particular stochastic mixture of I and Z.

For a resource family with defect-set decomposition

$$\rho = \sum_B \pi(B) \left(\bigotimes_{i \notin B} |R_i\rangle\langle R_i| \right) \otimes \sigma_{B,E}, \quad \sum_{B \supseteq J} \pi(B) \le q^{|J|}, \tag{D4}$$

a bad output requires at least two defective slots, in fact at distinct coordinates. For any u disjoint output runs, choose a pair in each; their $2u$ slots are distinct. The joint defect bound and union bound give

$$\Pr(\text{all } u \text{ outputs bad}) \le \binom{30}{2}^u q^{2u}. \tag{D5}$$

This proves Eq. (15) for the entire output family. The constant 435 is a conservative slot-pair count. It is consistent with the 21 and 105 distinct-coordinate pairs for the two codes and does not claim the cubic error suppression of postselected distillation.

Let y_ℓ, a_ℓ count raw candidates needed for one Y or A output at level ℓ. With preassigned correction slots,

$$y_\ell = 7y_{\ell-1}, \qquad a_\ell = 15a_{\ell-1} + 15y_{\ell-1}, \qquad y_0 = a_0 = 1. \tag{D6}$$

Since $7/15 < 1$, both are $O(15^\ell)$. The number 30 in the error bound therefore does not change the production exponent to $\log_2 30$. These recurrences count ideal logical factory operations up to a fixed factor; physical faults in those operations are addressed by the scheduled-circuit hypothesis.

Appendix E: Conditional threshold and resource accounting

This appendix proves the architecture implication under Assumption IV.1. It does not use finite certificates as proofs of that assumption. Let z bound the degree of the fixed union fault graph. The number of connected sets of m vertices containing a specified anchor is at most C_z^m for a size-independent constant C_z. One can see this by encoding a rooted spanning tree and its bounded-degree neighbor choices; overcounting suffices.

If at least a fraction θ of the vertices in such a set are actual faults, the local-stochastic condition and a union bound over their subsets give probability at most $2^m p^{\theta m}$. Subsets with incompatible labels at one location have probability zero, so they do not weaken the bound. Hence a large-cluster event over V possible anchors is bounded by a geometric sum of $V(2C_z p^\theta)^m$. Choosing a fixed threshold makes this sum exponentially small in its minimum size. Item 4 of the assumption supplies the required size and density properties for committed decoding; they are not consequences of bounded graph degree alone.

For clarity, the half-density statement in a fixed nonadaptive linear problem follows from minimum cardinality. If e is the actual error and $\widehat{e}$ a minimum solution with the same syndrome, then $e + \widehat{e}$ is detector silent. Each of its connected components is silent. Replacing the decoder's part on one component by the actual part remains a valid solution, so the actual part is at least as large as the decoder part there. At least half the residual component is therefore actual. Open commitment boundaries, mutually exclusive location labels, and changes of adaptive branch require the corresponding uniform witness statement rather than an unqualified extension of this argument.

1. Joint courier error estimate

Take a fixed-factor sequence $\lambda_- d_j \le d_{j+1} \le \lambda_+ d_j$ with $1 < \lambda_- \le \lambda_+$, and $R_j = \Theta(d_j)$. Set $u = C\sqrt{p} < 1$. The anchor and half-density bounds imply a one-resource charging estimate

$$q_{\text{inj}} \le C_0 u + C_1 \sum_{j=0}^{J} d_j^{5/2} u^{\gamma d_j} + C_1 f^{5/2} u^{\gamma d_J}. \tag{E1}$$

The first term is the raw stage. Choose the constant starting distance with $\gamma d_0 \ge 1$. For u below a fixed constant, the geometric growth of d_j makes the sum $O(u)$ uniformly in J. Choose

$$d_J \ge d_{\text{base}} + C_{\text{chain}} \frac{\log f}{\log(1/p)} \tag{E2}$$

with a sufficiently large fixed coefficient. Since $\log(1/u)/\log(1/p)$ is bounded below for sufficiently small p, the last term is also $O(\sqrt{p})$. The terminal block can remain below the full distance $\Theta(f)$, after adjusting constants and the minimum size, because $\log f = O(f)$.

For a specified set of distinct bad resources, item 4 supplies disjoint witnesses outside the large-cluster event. Choosing one witness and an actual-fault subset for each resource gives disjoint specified sets of actual locations. The same local-stochastic bound now applies to their union, and the witness-counting sums factor as upper bounds even though the noise is correlated. Thus the joint probability is bounded by $q_{\text{inj}}^{|J|}$. If the complement of the large-cluster event has probability at least $1/2$, conditioning costs at most a factor two, which is bounded by $2^{|J|}$ for nonempty J. Hence the conditional resource family has $q_0 \le 2q_{\text{inj}} = O(\sqrt{p})$. A marginal bound alone would not justify this conclusion.

2. Distillation levels and total failure

Writing $C_D = 435$, the ideal recursion gives

$$q_\ell \le C_D^{-1} (C_D q_0)^{2^\ell}. \tag{E3}$$

Choose a fixed $p_{\rm th}$ small enough that $C_D q_0 < 1$ uniformly below threshold. Taking $2^\ell = \Theta(f)$ makes the resource error exponentially small on the same scale as the protected-cluster error, and $15^\ell = O(f^{\log_2 15})$. The number of locations and consumed resources is at most WT times a fixed polynomial in f and 15^ℓ. Its polynomial factors can be absorbed by increasing the constant c and minimum value $f_{\min}$ in Eq. (17). A union bound allocates at most $\varepsilon/2$ to large clusters and at most $\varepsilon/2$ to final defective resources. This establishes the claimed implementation-failure bound under the uniform hypotheses.

Code parameters give $K = \Theta(\sqrt{f})$ and $n = \Theta(f^{3/2})$. A coarse logical box of side $\Theta(f^{1/6})$ fits into one block with fixed capacity reserved for live factory work. The number of blocks is $O(W/K + B_f(A) + 1)$, proving Eq. (18). Depth-first evaluation of a fixed-branching factory tree stores $O(\ell)$ candidates at once; $\ell = O(\log f)$ fits within the reserved $\Theta(\sqrt{f})$ capacity. The geometrically increasing courier sizes have bounded total volume relative to the largest one. Serial resource production then costs $O(Kf15^\ell)$ per logical layer, proving Eq. (19).

For a bounded-aspect logical box, with $x = W/\sqrt{f}$, $B_f(A) = O(x^{2/3} + 1)$. Since $x^{2/3} \le x + 1$, multiplication by $f^{3/2}$ gives $O(Wf + f^{3/2})$. Connectedness alone is insufficient for this bound: thin or fragmented regions can have many partially occupied coarse boxes. The space and time conclusions use a serial nearest-neighbor schedule with the specified hypotheses; they are not an unconditional threshold proof for every code deformation sharing the same static geometry.

Appendix F: Numerical records and reproducibility

The accompanying archive contains the revised `manuscript.tex`, `main_body.tex`, and `appendices.tex`, plus a `research_records` directory copied from the supplied author package. Paths in this appendix are relative to that directory. The main numerical sources are as follows.

- `two_block_distance.csv` and `two_block_mc.csv`: initial interface distances, solver statuses, and simulation counts. The implemented circuits and solver are in `two_block_measurement.py`.
- `numerical/scaling_campaign/mc_batches.csv`: unique task/batch records, seeds, shot counts, any-observable failures, and per-observable failures. The scheduler and profile are `scaling_campaign/campaign/run_mc.py` and `scaling_campaign/campaign/profiles.py`.
- `scaling_campaign/campaign/analyze.py` and `make_scaling_figure.py`: aggregation and suppression fits; `numerical/scaling_campaign/decoder_check.csv`: paired decoder comparison; the same numerical directory contains code and fault-distance results and the reported-bit witness checks.
- `numerical/continuous_transition/`: surface-code transition campaign, circuit generator, counts, and posterior crossing analysis. Headline runs use `final_transition_heavy.py`; posterior analysis is in `threshold_estimate/fit_threshold_estimate.py`.
- `growing_family/` and `routed_campaign_check.py`: routed-input campaign and its audit.
- `layer_code_explicit.py`, `slab_gluing_certificate.py`, `factory_map_audit.py`, and `boundary_composition_check.py`: finite construction, resource, and boundary-map checks. `schedule_verifier.py` checks a finite coarse schedule; it does not prove a uniform microscopic detector catalog.

TABLE V. Any-observable failures/shots for the spanning-grid suppression fits in Fig. 5(c). Counts aggregate unique archived batches. A dash denotes no data; zero-failure points are shown here but excluded by the archived fitting rule.

Basis	p (10^{-3})	$s=2$	$s=3$	$s=4$	$s=5$
X	0.5	52/68000	4/73200	0/25500	0/9900
X	1	257/58000	29/40800	7/22200	0/9000
X	2	49/2000	24/3000	25/6600	0/600
X	3	135/2000	135/4800	45/2100	5/300
X	4	289/2000	42/600	–	–
X	6	802/2000	180/600	110/300	–
Z	0.5	25/196000	1/72600	0/23400	0/9450
Z	1	29/34000	18/69000	3/22800	1/9300
Z	2	216/38000	31/14400	22/13500	1/1200
Z	3	32/2000	55/4200	35/5100	0/150
Z	4	90/2000	139/5400	65/2700	–
Z	6	322/2000	107/600	47/300	–

The revised numerical wording preserves archived observations and solver statuses. No inference of an exact distance is made from a FEASIBLE solver status alone. The independent revision checks reconstruct four complete small layer-code cases, verify their ranks and commutation, check 736 kernel-vector slab lifts at all positions, and check the teleportation identities and scalar arithmetic. Their script and results are supplied separately within the archive. These checks do not repeat the full Monte Carlo campaigns or certify the architecture hypotheses.

REFERENCES

[1] D. J. Williamson and N. Baspin, Layer codes, Nat. Commun. **15**, 9528 (2024), arXiv:2309.16503.

[2] D. J. Williamson and T. J. Yoder, Low-overhead fault-tolerant quantum computation by gauging logical operators, Nat. Phys. **22**, 598 (2026), arXiv:2410.02213.

[3] A. Cowtan, Z. He, D. J. Williamson, and T. J. Yoder, Parallel logical measurements via quantum code surgery, PRX Quantum **7**, 020325 (2026), doi:10.1103/gj8x-n5gg.

[4] A. C. Yuan, A. Cowtan, Z. He, T.-C. Lin, and D. J. Williamson, Parsimonious quantum low-density parity-check code surgery, arXiv:2603.05082 (2026).

[5] B. Ide, M. G. Gowda, P. J. Nadkarni, and G. Dauphinais, Fault-tolerant logical measurements via homological measurement, Phys. Rev. X **15**, 021088 (2025), doi:10.1103/PhysRevX.15.021088.

[6] L. Berent, L. Z. Cohen, and A. O. Quintavalle, Lifted surgery: Fast processing with QLDPC codes, arXiv:2609.11723 (2026).

[7] E. Swaroop, T. Jochym-O'Connor, and T. J. Yoder, Universal adapters between quantum low-density parity-check codes, PRX Quantum **7**, 010324 (2026), arXiv:2410.03628.

[8] Z. He, A. Cowtan, D. J. Williamson, and T. J. Yoder, Extractors: qLDPC architectures for efficient Pauli-based computation, arXiv:2503.10390 (2025).

[9] G. Zhang, Y. Zhu, and Y. Li, Accelerating fault-tolerant quantum computation with good quantum low-density parity-check codes, PRX Quantum **7**, 033012 (2026), doi:10.1103/3fj5-gn1h.

[10] A. J. Malcolm, A. N. Glaudell, P. Fuentes, *et al.*, Computing efficiently in QLDPC codes, Nat. Commun. **17**, 7286 (2026), doi:10.1038/s41467-026-73061-9.

[11] S. Bravyi, D. Poulin, and B. Terhal, Tradeoffs for reliable quantum information storage in 2D systems, Phys. Rev. Lett. **104**, 050503 (2010), arXiv:0909.5200.

[12] D. Horsman, A. G. Fowler, S. Devitt, and R. Van Meter, Surface code quantum computing by lattice surgery, New J. Phys. **14**, 123011 (2012), arXiv:1111.4022.

[13] D. Litinski, A game of surface codes: large-scale quantum computing with lattice surgery, Quantum **3**, 128 (2019), arXiv:1808.02892.

[14] M. Vasmer and D. E. Browne, Three-dimensional surface codes: Transversal gates and fault-tolerant architectures, Phys. Rev. A **100**, 012312 (2019), arXiv:1801.04255.

[15] C. A. Pattison, A. Krishna, and J. Preskill, Hierarchical memories: Simulating quantum LDPC codes with local gates, Quantum **9**, 1728 (2025), arXiv:2303.04798.

[16] S. H. Choe and R. König, How to fault-tolerantly realize any quantum circuit with local operations, arXiv:2402.13863 (2024).

[17] N. Baspin and D. J. Williamson, Wire codes, Quantum **10**, 2083 (2026), arXiv:2410.10194.

[18] Y. Yang, G. Zhang, and Y. Li, Planar fault-tolerant logical measurements with low qubit overhead, npj Quantum Inf. (2026), doi:10.1038/s41534-026-01310-2, arXiv:2506.18061.

[19] D. Gottesman, Fault-tolerant quantum computation with constant overhead, Quantum Inf. Comput. **14**, 1338 (2014), arXiv:1310.2984.

[20] A. A. Kovalev and L. P. Pryadko, Fault tolerance of quantum low-density parity check codes with sublinear distance scaling, Phys. Rev. A **87**, 020304(R) (2013).

[21] O. Fawzi, A. Grospellier, and A. Leverrier, Constant overhead quantum fault tolerance with quantum expander codes, in *FOCS 2018*, pp. 743–754 (2018), arXiv:1808.03821.

[22] G. Zhang, Y. Zhu, X. Yuan, and Y. Li, Constant-overhead magic state injection into qLDPC codes with error independence guarantees, arXiv:2505.06981 (2025).

[23] K. Liu, S. Xu, T. Jochym-O'Connor, Z. He, S. Singh, and Y. Ding, In-situ simultaneous magic state injection on arbitrary CSS qLDPC codes, arXiv:2604.05126 (2026).

[24] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A **71**, 022316 (2005), arXiv:quant-ph/0403025.

[25] B. W. Reichardt, Quantum universality from magic states distillation applied to CSS codes, Quantum Inf. Process. **4**, 251 (2005).

[26] D. Litinski, Magic state distillation: not as costly as you think, Quantum **3**, 205 (2019).

[27] A. Leverrier and G. Zémor, Quantum Tanner codes, in *2022 IEEE 63rd Annual Symposium on Foundations of Computer Science (FOCS)*, pp. 872–883 (2022), doi:10.1109/FOCS54457.2022.00117, arXiv:2202.13641.

[28] I. Dinur, S. Evra, R. Livne, A. Lubotzky, and S. Mozes, Locally testable codes with constant rate, distance, and locality, in *Proceedings of the 54th Annual ACM SIGACT Symposium on Theory of Computing (STOC '22)*, pp. 357–374 (2022), doi:10.1145/3519935.3520024, arXiv:2111.04808.

[29] T. Jochym-O'Connor, A. Kubica, and T. J. Yoder, Disjointness of stabilizer codes and limitations on fault-tolerant logical gates, Phys. Rev. X **8**, 021047 (2018), doi:10.1103/PhysRevX.8.021047, arXiv:1710.07256.

[30] S. Gu, L. Caha, S. H. Choe, Z. He, A. Kubica, and E. Tang, Layer codes as partially self-correcting quantum memories, Phys. Rev. Lett. **136**, 210601 (2026), doi:10.1103/x4z2-f4gg, arXiv:2510.06659.